

Marriage and Misallocation: Evidence from 70 Years of U.S. History*

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Abstract

The traditional expectation that married women should be homemakers restricts them from pursuing their comparative advantage in the labor market. I quantify the aggregate economic consequences of these marriage-specific gender norms, accounting for selection into marriage and labor force participation. In 1940, married women faced a “norms wedge” equivalent to a 44% tax on market wages compared to similar single women, which fell to 14% by 2010. Had these norms persisted at 1940 levels, market output today would be 8.7% lower and combined market and home output 5.3% lower. Amplification effects through endogenous human capital investment and marriage decisions, which reshape the productivity distribution of both women and men, are critical to generating these large results.

Keywords: Marriage, gender norms, misallocation, female labor force participation

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1 Introduction

Despite converging economic lives of men and women – in time use, education, and earnings – in many countries, substantial gender specialization remains between married couples.¹ Under certain forms of the household production function, specialization is required for efficiency (Becker, 1981; Pollak, 2013) and would therefore benefit aggregate productivity. This paper focuses on an alternative view of specialization: to the extent that married women engage in home production to fulfill the traditional gender role of the homemaker, some may be prevented from pursuing their comparative advantage in market work, resulting in misallocation.

By how much do traditional gender norms in marriage constrain aggregate output? This paper quantifies the effect on aggregate output of changes in the “homemaker” gender role in the U.S. between 1940 and 2010, within a general equilibrium model of education, marriage, and labor supply. The key innovation is my focus on gender roles within marriage, where women become subject to these norms only upon marrying. This differs fundamentally from existing work that examines aggregate productivity effects of gendered institutions that focus on the inherent difference between men and women (Hsieh et al., 2019; Erosa et al., 2022; Lee, 2024; Chiplunkar and Goldberg, 2024). In this paper, whether a woman faces traditional gender roles depends on her marriage status, an endogenous choice. This endogeneity makes selection into marriage a central consideration.

Because marriage is endogenous, the general equilibrium effects of gender norms are far from straightforward. In partial equilibrium – holding individuals’ education and marital status fixed – the implication is simple: traditional norms lead married women to work less. In general equilibrium, however, these norms also influence decisions around both marriage and education, making their overall impact more complex. For instance, the effect of conservative norms on women’s labor-market incentives to get educated depends on how those norms shape expectations about future labor force participation at different levels of education. If stricter norms discourage labor market participation among married women but also reduce the likelihood of marriage, particularly among highly educated women, the net effect on women’s labor-market returns to education becomes ambiguous. Moreover, the impact on educational attainment is further complicated by the presence of marriage-market returns to education, which may reinforce or offset labor-market considerations.²

Individuals in my model make three sets of choices over the course of their life cycle. First,

¹Lundberg and Pollak (2007) review this pattern for the United States. Gershuny (2018) document a consistent within-country gender convergence in time use, with men doing more unpaid work and women more paid work, in 12 European countries, South Korea, the United States, and Australia from the 1960s to the 2010s; yet, women still perform nearly 60% of unpaid work even where convergence is most advanced.

²A more detailed description of the channels through which changes in gender norms affect aggregate productivity is in Section 3.7.

individuals choose their level of education as a forward-looking investment, considering both labor- and marriage-market returns. Second, they enter a frictionless transferable-utility marriage market in the style of [Becker \(1973\)](#), where individual types are defined by education. They decide which spousal type to marry or to stay single, and then draw a children composition category (i.e., number and ages of children) according to match-specific empirical probabilities. Fertility is not modeled as a structural choice for tractability, as I always net out the effect of children to isolate marriage-specific gender norms.³ Third, individuals draw market and home abilities, and households make the dichotomous labor supply choice of whether to work in the market or on home production for each individual. I abstract from work hours or from distinguishing part-time and full-time work, because the growth in married women’s total market work hours between 1940 and 2010 is almost entirely driven by the extensive margin of participation rather than the intensive margin of hours conditional on working.⁴ Gender roles are modeled as a disutility for a married couple when the wife works in the market ([Akerlof and Kranton, 2000](#); [Bertrand et al., 2015](#)), represented by a “norms wedge” that acts like a tax lowering the value of her market wage.

The model is matched to the education, marriage, and labor force participation patterns in the U.S. decennial census, decade by decade, to track how the magnitude of norms wedges changes over time. Norms wedges are calibrated as a composite force that makes the labor force participation of married women diverge from that of similar⁵ single women besides the wage differentials, encompassing many factors that fall under the broad concept of homemaker norms.^{6,7} I also address major disparities between married and single women’s work decisions unrelated to norms, including income effects from husbands’ earnings, tax advantages for marriage ([Guner et al., 2012](#); [Gayle and Shephard, 2019](#)), and differing childbearing prospects. As validation, I confirm that the model-derived measures of norms are highly correlated with attitudinal survey responses, both across U.S. states and across time within states. With the calibrated model, I simulate counterfactual scenarios to examine how changes to the norms wedge directly affect labor supply choice, indirectly influence

³I net out the effect of children when comparing married and single women in the motivating evidence (Figures 1 and 2) and in the calculation of the norms wedge. Nonetheless, the match-specific empirical probabilities of children composition are calibrated separately for each decade, allowing the model to capture shifts in fertility patterns – such as the baby boom and subsequent decline, and changes in how fertility relates to income, female labor force participation, and female education ([Doepke et al., 2023](#)).

⁴See Appendix C for a visual illustration and a formal decomposition.

⁵I.e., matched on education and children composition (e.g., university education and one child under 5).

⁶For example, norms wedges encompass the preference to conform to traditional gender norms as a wife, differential valuation of staying home for married women relative to single women, differential non-wage treatment of married women by firms relative to single women, and men’s preference to have more home-productive women as wives, etc.

⁷Throughout this paper, I treat cohabiting couples as married, since they are subject to similar – albeit likely weaker – gender roles as legally married couples. I define married couples as those with “probable” spouses living in the same household, as identified in the IPUMS census data ([Ruggles et al., 2024](#)), which includes cohabiting couples from 1990. Appendix G.2 examines the sensitivity of the calibrated norm wedges to the definition of marriage by legal marital status only; the wedges are slightly higher, as expected.

marriage and educational choice in anticipation, and ultimately, aggregate output.

I find that married women faced a 44% norms wedge on the market wage in 1940, which declined to 14-15% by 2000 and 2010. If gender norms had stayed at the level of 1940, aggregate market output in 2000 would be lower by 8.7% and aggregate total (market and nonmarket) output would be lower by 5.3%. These drops represent 12.6% of the growth in market output and 8.1% of the growth in total output between 1940 and 2000. To gauge the magnitude of these results, I compare them to the output effect of another counterfactual: if married women of 2000 made labor force participation choices based on 1940 wages and home productivities. I find that distortions driven by cultural forces are almost 70% as consequential as those driven by economic forces. I also explore counterfactual output under alternative modeling assumptions and find that norms wedges continue to generate large distortions even in conservative setups.

It is the double forces of lower human capital in the aggregate and greater misallocation of the market talent of married women that drive the output effects. Importantly, a large part of the overall output decline stems from falling market productivity of men, which would be missed in a partial equilibrium analysis that holds education and marriage choices fixed. Average years of schooling drop by 2.0% for men and 2.6% for women. This is partly the result of marriage becoming less attractive and thus the marriage-market returns to education falling. The marriage rate indeed falls by a quarter, especially among highly educated women. For women, schooling decisions also respond to labor-market returns, which are found to reinforce the decline in education caused by weaker marriage incentives. Married women's labor force participation and cumulative market earnings drop by 34.1% and 28.2%, respectively, reflecting both stronger misallocation with higher norms wedges and the more negative selection into marriage. Furthermore, changes in marriage patterns interacts with education-driven productivity losses in asymmetric ways: because married men earn more than single men, the drop in marriage amplifies the loss in men's market productivity; for women, in contrast, married women earn less than single women, so lower marriage rates mitigates their productivity loss.⁸ Consequently, latent market productivity of men falls by 7.2%, exceeding the 2.9% decline for women.

The counterfactual also implies that the reduction in the gender norms wedges between 1940 and 2010 accounts for several well-documented trends in the U.S.: a) rising married female labor force participation rate, b) increase in wife's share of household income, c) growing educational attainment of both men and women, but especially women, and d) increasingly positive selection into marriage by education ([Greenwood et al., 2016](#); [Bar et al., 2018](#)). In particular, it accounts for 38% of the rise in the labor force participation of married women.

⁸The difference in married and single wages reflect occupational choices, firm sorting, job mobility, hours, work schedule flexibility, and work effort – shaped directly by marital status and indirectly by differential likelihood of having children. As a conservative robustness check, I test the sensitivity of my counterfactual results to assuming that marriage has no causal effect on productivity.

The analysis in this paper is based on historical data from the United States. Yet, numerous countries in the world are experiencing similar trends as the U.S. had: gender attitudes have become less conservative,⁹ and married women’s labor force participation is catching up with single women’s. These countries include not only the richer OECD countries but also low- and middle-income countries in Eastern Europe and Latin America. At the same time, one in ten countries of the world still has a lower female labor force participation rate than 1940 U.S. ([International Labor Organization, 2019](#)).¹⁰ Meanwhile, in the United States, small but increasingly visible groups – such as religious traditionalists, Christian nationalists ([Whitehead and Perry, 2020](#)), and certain online communities ([Ging, 2019](#)) – are advocating a return to stricter gender norms that discourage married women’s labor force participation.¹¹ Thus, this paper informs us about the potential growth consequences and the underlying channels of cultural change in countries currently transitioning away from traditional gender roles, as well as what may be at stake should such norms regain traction.

A large literature pioneered by [Restuccia and Rogerson \(2008\)](#) and [Hsieh and Klenow \(2009\)](#) study the aggregate productivity implications of various forms of misallocation. A growing number of papers examine gender-based distortions in the allocation of talent – discrimination in the educational and labor markets distorting occupational choice ([Hsieh et al., 2019](#)); differences in nonmarket time using married couples only, without considering marriage as a choice ([Erosa et al., 2022](#)); entry barriers to the non-agricultural sector ([Lee, 2024](#)); and barriers to entrepreneurship ([Chiplunkar and Goldberg, 2024](#)). I add to this literature by focusing primarily on the distinction between *married and single women*, rather than comparing men and women. By integrating the marriage market matching into the model, I explore a new set of channels behind the aggregate productivity effects, such as selection into marriage and marriage-market returns to education.

I also contribute to the literature at the intersection of macroeconomics and family economics, which has placed emphasis on how marriage and family formation are critical for understanding various aggregate outcomes – business cycles, income inequality, fertility, labor supply, savings, long-run development, and political institutions (see [Doepke and Tertilt \(2016\)](#) for an overview). Yet, little is known about how gender norms tied to marriage shape aggregate output. This paper fills that gap.

My theoretical contribution is to build a tractable general equilibrium model with closed-form

⁹Conservativeness in gender attitudes is measured by the fraction agreeing to “When jobs are scarce, men have more right to a job than women,” asked in the World Values Survey.

¹⁰In light of the U-shaped pattern of female labor force participation over the cycle of economic development ([Goldin, 1995](#); [Mammen and Paxson, 2000](#); [Ngai et al., 2024](#)), found when unpaid family labor is included, I should clarify that the mentioned ILO statistic pertains only to paid employment and self-employment.

¹¹Their influence may exceed their actual size due to a documented tendency to misperceive prevailing social norms by overweighting minority views on whether “women should have the freedom to work outside the home” in the United States and many other countries ([Bursztyrn et al., 2023](#)).

solutions, while maintaining flexibility to account for key secular trends in the gender wage gap, gender differences in home productivity, propensity of marriage, assortativeness of marriage matching by education, fertility patterns, and educational attainment. The novelty lies in extending a tractable selection framework – commonly used in the trade literature (Eaton and Kortum, 2004; Hsieh et al., 2019) to study *individual* decisions – to *household* decision-making, and integrating it with standard models of educational choice and marriage market matching (Choo and Siow, 2006; Chiappori et al., 2017; Chiappori et al., 2018). The resulting closed-form solutions clarify how education, marriage, and labor supply decisions interact and make the parameter identification procedure fully transparent. While recent structural work on marriage markets and family labor supply (e.g., Goussé et al., 2017; Gayle and Shephard, 2019; Calvo et al., 2024) incorporates richer mechanisms, estimating a large number of parameters and comparing them consistently across many time periods can be challenging due to their complexity. In contrast, the combination of tractability and flexibility in my framework is particularly well-suited for tracing the evolution of norms wedges decade by decade over 70 years.

Lastly, I contribute to a large body of both macro and micro work that seeks to explain the dramatic rise in married women’s labor force participation in the U.S. Existing explanations can be broadly categorized into two branches: technological progress and cultural change. The “technology” branch includes the invention of birth control pills (Goldin and Katz, 2002), technological advances in housework (Greenwood et al., 2005), gender-biased technological progress (Heathcote et al., 2010), medical progress in pregnancy-related conditions (Albanesi and Olivetti, 2016), and the rise in services driven by structural transformation and marketization (Ngai and Petrongolo, 2017). The “culture” branch includes greater acceptance of working wives by men (Fernández et al., 2004), changes in social attitudes toward married women working (Fernández, 2013), reforms to divorce laws (Fernández and Wong, 2014), and falling gender wage discrimination (Jones et al., 2015). I add to the latter branch by providing a model-driven way to measure *marriage-specific* gender norms – an understudied dimension of cultural constraints – and quantifying their aggregate importance as a driver of the rise in married women’s labor force participation over the last several decades.

The remainder of the paper is organized as follows. Section 2 presents empirical evidence motivating the focus on marriage-specific norms that differentiate the labor supply of married and single women. Section 3 develops a model that formalizes these differences and explores their implications for education and marriage decisions. Section 4 describes the data, outlines the calibration procedure, and discusses the resulting parameter estimates. Section 5 uses the model to quantify the impact of changes in gender norms on aggregate productivity. Section 6 concludes.

2 Motivating Facts

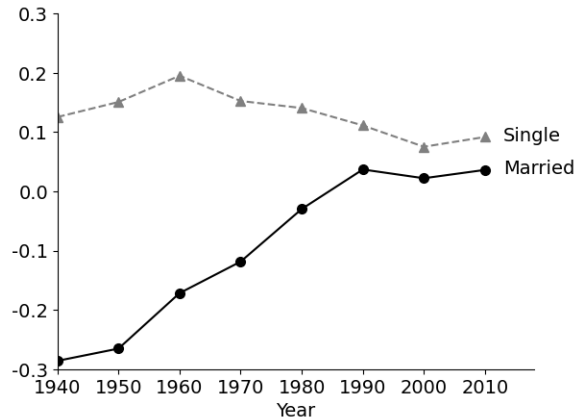
This section presents descriptive facts that motivate my focus on traditional gender roles as the key distinguishing factor between married and single women’s labor supply, and examines how gender roles have evolved over time.

“Unexplained” female labor force participation I compare the path of the “unexplained” labor force participation (LFP) of married and single women over time. By unexplained LFP, I refer to residuals from regressing the LFP indicator on standard, commonly observed individual characteristics X : age, education, race, and number of children dummies.

$$LFP_{it} = X_{it}\beta + \epsilon_{it}$$

After running the regression on all females aged 25-54 between 1940 and 2010 in the U.S. decennial census, I take the weighted average of the residuals by marital status and decade.

Figure 1: Residualized Female Labor Force Participation, by Marital Status



Note: This figure compares the labor force participation rates of single (never married, separated, divorced, widowed) to married women between the ages of 25 and 54 in 1940 and 2010, in the United States. The participation rates are residualized for age, education, race, and number of children dummies.

Figure 1 shows that unexplained LFP rose for married women over this period, catching up to the relatively stable level for single women. This highlights, firstly, that the LFP choices of married women differ substantially from those of single women, and secondly, that this difference has diminished over time. Importantly, this distinction persists even when controlling for the number of children, demonstrating that differences in labor supply behavior between married and single women extend beyond the presence of children – the dominant factor setting marrieds apart from singles in the literature. Hence, technological advances in child-bearing or child-rearing cannot

explain all of the catch-up, leaving room for cultural change around gender roles within marriage as a potential contributor.

Shift in work patterns upon marriage To further corroborate the observation that married women’s labor supply choice is disparate from single women’s, I analyze how women adjust their time use immediately upon marriage. I contrast these shifts with those observed among men to shed light on gender roles within marriage.

I follow the event-study approach of [Kleven et al. \(2019\)](#). For this exercise, I use the Panel Survey of Income Dynamics 1968-2015 ([Panel Study of Income Dynamics, public use dataset, 2024](#)), a panel data of nationally representative individuals of the United States. Men and women often live in different family arrangements prior to marriage – for example, women may be more likely to reside in their parental home, while men may be more likely to live alone. In addition, some couples cohabit before marrying. To minimize the influence of other adult household members or cohabiting partners on time use before marriage, I restrict the sample to individuals who transition from being unmarried household heads without any live-in partners, to being married with live-in spouses.¹²

I define event times to be years relative to marriage, such that event time 0 refers to the first year in which an individual’s marital status switches to being married from being single. I run the regression

$$Y_{ist} = \sum_{j \neq -1} \alpha_j \cdot \mathbb{1}(t = j) + \sum_k \beta_k \cdot \mathbb{1}(age_{is} = k) + \sum_y \gamma_y \cdot \mathbb{1}(s = y) + X_{ist}\delta + \nu_{ist}$$

This regression tracks how the outcome variable Y of individual i in calendar year s at event time t changes as a function of event time, while controlling for life-cycle dynamics via the age dummies and time trends via the year dummies. I consider two outcome variables: labor force participation and weekly housework hours.¹³ X includes additional control variables – the number of children and the age of the youngest child – to distinguish the evolution of outcomes around marriage from that around childbirth. Regressions are run separately by gender and, to examine whether and how “marriage effects” change over time, across two distinct marriage cohorts: earlier and later marriages.¹⁴

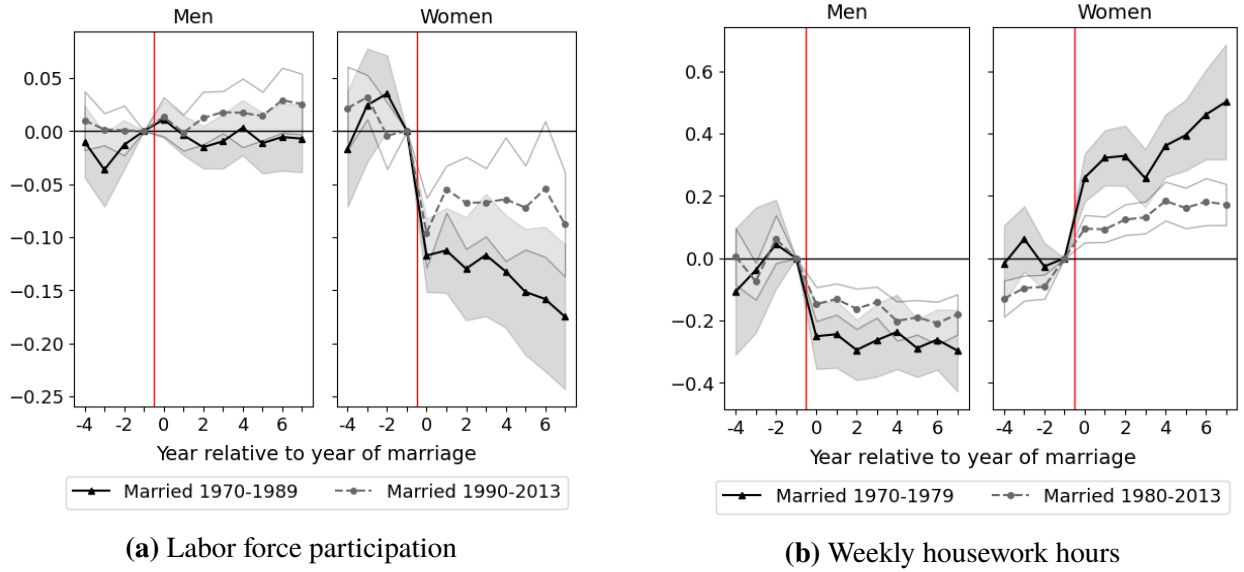
¹²Additional sample restrictions are: (i) only including years from three years before marriage (the year marital status changes from 0 to 1) to eight years after, (ii) individuals with only one marriage recorded for a clean event study design, and (iii) restricting marriages to those occurring between 1970 and 2013.

¹³For housework hours, the survey asks, “About how much time does (he/she) (do you) spend on this housework in an average week—I mean time spent cooking, cleaning, and other work around the house?”

¹⁴For housework hours – measured from 1969 – I define marriage cohorts as individuals married 1970–1979 (early) and 1980–2013 (late). In contrast, employment status of wives of household heads is only measured from 1979. To ensure sufficient pre-marriage data for the early cohort when analyzing labor force participation, I define the marriage cohorts as those married 1970–1989 (early) and 1990–2013 (late).

The event time coefficients ($\hat{\alpha}_t$) are then normalized by $\mathbb{E}[\tilde{Y}_{ist}|t]$, where $\tilde{Y}_{ist} \equiv \sum_k \hat{\beta}_k \cdot \mathbb{1}(age_{is} = k) + \sum_y \hat{\gamma}_y \cdot \mathbb{1}(s = y) + X_{ist} \hat{\delta}$ is the predicted outcome when excluding the effect of the event time. $\hat{\alpha}_t / \mathbb{E}[\tilde{Y}_{ist}|t]$ are plotted in Figure 2.

Figure 2: Event Study – Labor Force Participation and Housework Hours Around the Year of Marriage



Note: This figure plots $\hat{\alpha}_t / \mathbb{E}[\tilde{Y}_{ist}|t]$, the evolution of Y around the year of marriage relative to its predicted level in the absence of marriage. The red vertical line plots the timing of marriage. Individuals are unmarried household heads without any live-in partners in the years to the left of the red line, and they are married with live-in spouses in the years to the right of the red line. 95% confidence intervals are reported, based on bootstrapped standard errors computed using 100 resamples at the individual level to account for the panel structure of the data.

Panel (a) plots the results for labor force participation. It illustrates a sharp drop in women's labor force participation immediately upon marriage, and more so for the earlier marriage cohort (-11.7% drop relative to the predicted level in the absence of marriage) than the later one (-9.6%). In contrast, there are no sharp changes for men in either marriage cohort. Panel (b) plots the results for weekly housework hours. There is a sharp 22.6% increase in women's weekly housework hours immediately upon marriage for the earlier marriage cohort, whereas the increase for the later cohort is more muted. For men, there is a sharp decrease upon marriage that is greater for the earlier cohort.

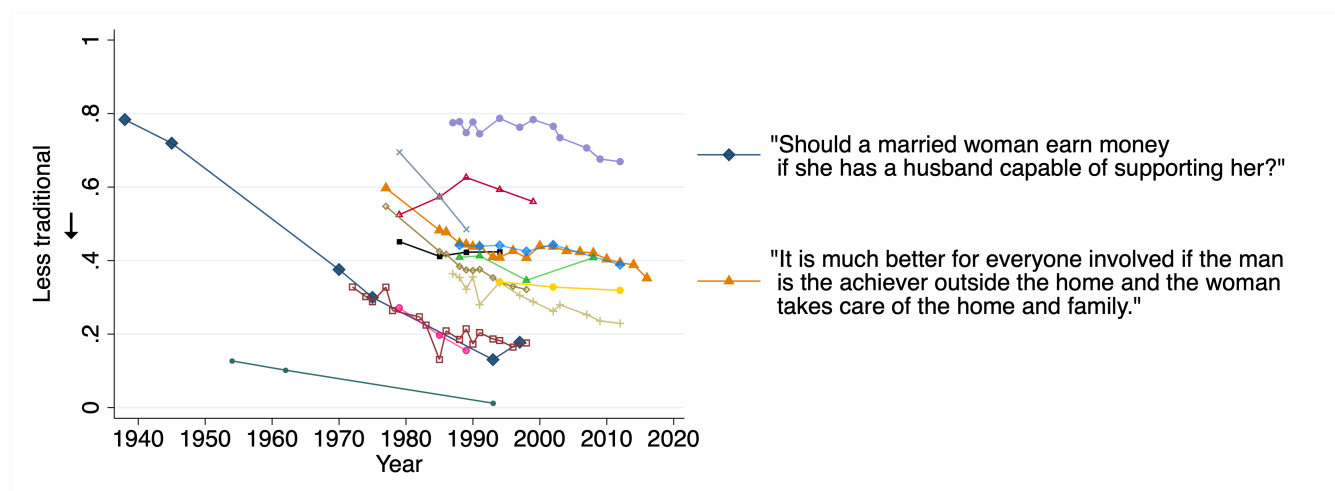
The stark divergence in labor force participation and housework hours between men and women upon marriage shows that marriage plays a pivotal role in shaping gendered divisions of labor, particularly in earlier marriage cohorts. Since the regression controls for the number of children and the age of the youngest child,¹⁵ the results further suggests that marriage itself – independent

¹⁵The regression results remain similar if instead of controlling for these, I restrict the sample to couples who had

of parenthood – imposes gendered expectations on women. This interpretation aligns with the finding in Figure 1, where married women’s labor force participation remains lower than that of single women, even after adjusting for the number of children. The attenuated shifts observed for women in the later marriage cohort may partly reflect the increasing marketization of household chores over time – such as the greater availability of household appliances (Greenwood et al., 2016). When combined with the smaller decline in housework hours for men in the later cohort, however, they also suggest a broader shift in social expectations.

Attitudinal survey trends The final motivating fact directly illustrates the weakening of gender roles within marriage over time, a pattern documented in prior work (e.g. Fernández (2013) for the U.S., 1938-1998; Goussé et al. (2017) for the UK, 1991-2008). To capture a more comprehensive view of attitudinal trends that covers every decade from 1940 to 2010, I compile attitudinal survey data from the ROPER polls database. From the set of survey questions returned by keyword searches for “women” and “gender,” I select those related to gender roles within marriage that were asked repeatedly over multiple time periods. For comparability across questions, responses are rescaled to lie between 0 and 1, with 1 indicating the most traditional answer and 0 the least.

Figure 3: Trend in Attitudinal Survey Answers



Note: This figure plots the yearly weighted mean of responses to attitudinal survey questions related to gender roles within marriage from the ROPER polls database. The precise survey questions are shown for two questions that have been asked for the longest periods and together span the entire time frame. A complete legend with all 14 survey questions is provided in Appendix Figure A1. All surveys are based on nationally representative samples of adults or, in some cases, of men and women separately.

Figure 3 shows a clear trend toward less traditional attitudes over time, though the pace of change may be slowing in more recent decades. The variation in the answers to these questions no childbirths in the first three years of marriage.

across states and decades is used in Section 4.3 to cross-check the validity of the model-derived measures of norms.

3 Model

The motivating facts presented in Section 2 reveal that labor supply decisions differ systematically between married and single women, with gender roles potentially accounting for this disparity. They also show that the gap has narrowed over time. In this section, I develop a model that formalizes how gender norms within marriage shape women’s labor supply choices. Because these norms specifically affect married women, it is essential to model selection into marriage. The model therefore incorporates marriage matching based on educational attainment, the primary observable trait along which couples sort and the key predictor of gender attitudes.¹⁶

Timing The decisions of individuals are divided into three stages. In the first stage, individuals choose their level of education as a forward-looking investment, balancing the returns to education in the labor and marriage markets with the cost of education, as in Chiappori et al. (2009).

In the second stage, they enter a frictionless transferable utility (TU) marriage market (Becker, 1973), where types of individuals equal their education levels chosen previously, and decide on which spousal type to get married to or to stay single. The resulting match is thus characterized by the education levels of the husband and wife for married couples and by one’s own education level for singles. Households then exogenously get assigned children composition categories based on the number of children under the age of 5 and under the age of 18 in the household, according to match-specific empirical probabilities. Fertility is not modeled as a structural choice for tractability, as I always net out the effect of children to isolate marriage-specific gender norms.

Individuals subsequently enter the third and last stage, each characterized by group: the tuple of (gender, marriage match, children composition). After individuals draw idiosyncratic market and home abilities, households make the dichotomous labor supply choice of whether to work in the market or in home production for each individual, taking as given a) the group-specific market wage that the representative firm of the economy pays, b) the group-specific value of home production, and c) the group-specific disutility that a married couple gets upon deviation from traditional gender

¹⁶The model does not include attitudes toward gender roles as part of individuals’ types in the marriage market. Consequently, it does not capture, for instance, whether more traditional men or women are more likely to marry, or whether individuals match based on shared family values. However, this type of matching appears empirically improbable. I show in Appendix Table A1 that attitudes toward gender roles are no different by marital status, using U.S. attitudinal survey data described in Section 2. Similarly, Goussé et al. (2017), who explicitly incorporate family values into individual types, find no evidence of homophily in family values in Britain between 1991 and 2008. In fact, they find no evidence of complementarities in any pairwise interaction of market wages (labor productivity), family values, and education of men and women – except for homophily in education.

roles, i.e. when the wife works in the market. I focus on the binary work decision because the growth in married women’s total market work hours between 1940 and 2010 is almost entirely driven by the extensive margin of participation rather than the intensive margin of hours conditional on working (see Appendix C). After the labor supply decisions are made, households consume and realize utilities.

Since I solve the model backwards, I describe each stage in greater detail, in reverse order.

3.1 Economic utilities and optimal labor supply choices

Individual utilities consist of an economic and a predetermined noneconomic component. The *economic* component is characterized by the utility functions below, adapted from Chiappori et al. (2018).¹⁷ The economic gains from marriage stem from two sources: economies of scale in consuming public goods and risk sharing between spouses against uncertain future public and private consumption.

Married Individuals Consider a married household composed of husband m and wife f . Individual $i \in \{m, f\}$ gets the following utility:

$$u_i(Q, C_i, L_f) = \ln(Q) + \ln(C_i - \tau_i w_f L_f) \quad (1)$$

where C_i is the consumption of the private good, and Q is the consumption of the public good (e.g. expenditures on housing, children, heating). $L_i \in \{0, 1\}$ denotes the labor market participation, and w_i the market wage, of individual i . Critically, married individual i gets disutility when the wife works in the market. The parameter τ_i measures the value of the disutility as a proportion of the market income brought home by the wife.¹⁸ The focus on gender norms affecting the wife’s LFP, but not the husband’s, is motivated by the absence of any sharp change in men’s LFP upon marriage – contrary to sharp drops for women – shown in the previous section (Figure 2).

From an ordinal perspective, this utility belongs to Bergstrom and Cornes (1983)’s Generalized Quasi Linear (GQL) family. Hence, for any realization of family income, it satisfies the transferable utility (TU) property. Under TU, utility can be transferred between spouses at a fixed rate of exchange, and so for Pareto-efficiency, a couple acts as a single decision unit that maximizes the joint marital output. As the set of Pareto efficient allocations is an ordinal concept, any cardinalization of u can be used for the definition of joint marital output. I use $\exp(u_i)$ as the cardinalization of i ’s preferences. Then, conditional on the couple’s labor choices, any Pareto efficient allocation

¹⁷A more general formulation of the utility function that is consistent with my model is described in Appendix D.1.

¹⁸The multiplicative relationship between τ_i and w_f implies that, for a given value of τ_i , the disutility is greater in absolute terms for households with higher-earning wives – i.e., norms around gender roles are most costly in these households. One rationale is that higher wages often correspond to longer work hours, and the disutility from violating gender norms increases with the time spent in market work.

maximizes the sum of the spouses' exponential utilities, $\exp(u_m) + \exp(u_f)$.

Therefore, conditional on labor market participation choices, a married couple solves

$$\begin{aligned} \max_{Q, C} & Q(C - \tau w_f L_f) \\ \text{s.t.} \quad & pQ + C = w_m L_m + w_f L_f + h_m(1 - L_m) + h_f(1 - L_f) \end{aligned} \quad (2)$$

$C \equiv C_m + C_f$ denotes total expenditure on private goods, $\tau \equiv \tau_m + \tau_f$ is the couple's joint disutility from the wife working in the market, and p is the price of the public good relative to the private good (the numeraire). Moreover, h_i refers to the home productivity of individual i .¹⁹ The solutions to the maximization problem given by (2) are

$$\begin{aligned} Q &= \frac{w_m L_m + (1 - \tau)w_f L_f + h_m(1 - L_m) + h_f(1 - L_f)}{2p} \\ C &= pQ + \tau w_f L_f \end{aligned} \quad (3)$$

I now describe the intra-household allocation, before market earnings and home productivities are realized. Efficient sharing of risks against the uncertainty of earnings and home productivities implies that the ratio of marginal utilities of private consumption is constant and equal to the Pareto weight (μ),²⁰ determined endogenously in the marriage market:

$$\frac{\partial u_m}{\partial C_m} = \mu \frac{\partial u_f}{\partial C_f}$$

The resulting indirect utilities are:

$$v_m = 2 \ln Q + \ln p + \ln \frac{1}{1 + \mu}, \quad v_f = 2 \ln Q + \ln p + \ln \frac{\mu}{1 + \mu} \quad (4)$$

The total economic utility generated from this marriage then is

$$v = v_m + v_f = \check{v} + \ln \frac{\mu}{(1 + \mu)^2} \quad (5)$$

where $\check{v} \equiv 4 \ln Q + 2 \ln p$. It is straightforward that the farther the wife's Pareto weight μ is from 1, the husband's, the smaller the total economic utility of the couple.

The optimal labor choices, which maximize Q , are

$$\begin{aligned} L_m^* &= \mathbb{1} [w_m \geq h_m] \\ L_f^* &= \mathbb{1} [(1 - \tau)w_f \geq h_f] \end{aligned} \quad (6)$$

In the labor choice of married women, τ enters as a “norms wedge”: a married woman's market

¹⁹It might seem odd that according to the budget constraint, private and public goods can be bought with “income” from home production. However, Appendix D.2 shows that the utility maximization problem remains identical if I divide goods into market goods and home-produced goods, and have two separate budget constraints for each.

²⁰Browning et al. (2014) characterize intra-household allocations under efficient risk-sharing.

wage is valued to be lower than its face value, as if it is taxed.²¹ One feature that stands out in solution (6) is the independence of the husband’s and wife’s labor supply choices.²² While this feature makes studying the selection into working in the labor market by either the husband or the wife easy, it is not realistic. I consider an extension of the model that I refer to as the “extension with income effects” in Appendix D.4: assuming men always work, I relax the independence so that the labor supply decisions of women are responsive to their husbands’ market earnings. I show later in Section 4.3 that the benchmark model with independent spousal decisions performs well up to 2000.

The flexibility of the model allows for another extension (Appendix D.5) where norms wedges apply also to married men, such that gender norms additionally impose costs on the household when the husband is a homemaker.²³

Single Individuals To distinguish from the married case, I use the hat symbol for singles. The economic utilities of singles follow the same structure as those of married couples, except that they are not subject to gender norms. Consequently, single women do not incur disutility from participating in the labor market.

A single individual i solves

$$\begin{aligned} \max_{Q_i, \hat{C}_i, \hat{L}_i} \quad & \hat{u}_i(\hat{Q}_i, \hat{C}_i) = \ln(\hat{Q}_i) + \ln(\hat{C}_i) \\ \text{s.t.} \quad & p\hat{Q}_i + \hat{C}_i = w_i\hat{L}_i + h_i(1 - \hat{L}_i) \end{aligned} \quad (7)$$

The resulting indirect utility is

$$\hat{v}_i = 2 \ln \hat{Q}_i + \ln p \quad (8)$$

and the optimal labor choice that maximizes $\hat{Q}_i$ is

$$\hat{L}_i^* = \mathbb{1} [w_i \geq h_i] \quad (9)$$

Market Wage and Home Productivity Market wages and home productivity depend on idiosyncratic market and home abilities, as well as components common to groups defined by gender, marriage match, and children composition. I abstract from fertility decisions, and instead assume that children composition is drawn exogenously according to match-specific empirical probabilities. Hence, the probability that a (husband education level q , wife education level r) match has

²¹The norms wedge functions as a catch-all term that encapsulates many different factors that make married women’s LFP diverge from single women’s LFP besides the wage differentials. See Section 4.3 for a discussion of what these factors include, and how they fall under the broad concept of homemaker norms for married women.

²²Relatedly, as $\frac{\partial^2 \ln Q}{\partial w_m \partial w_f} = 0$, the model does not build in any assortative matching on market earnings.

²³Appendix G.1 presents the calibration of these “breadwinner” norms wedges for married men. They decrease over time like their female counterparts. However, they come with some caveats, which I discuss.

a children composition $\mathcal{K}$, denoted as $d^{qr}(\mathcal{K})$, is simply found from corresponding shares in the data.²⁴

Individual i of gender g in a (q, r) match, with children composition $\mathcal{K}$, earns market wage

$$w_i = \bar{w}_g^{qr}(\mathcal{K})\epsilon_i^w, \quad g \in \{M, F\}$$

where $\bar{w}_g^{qr}(\mathcal{K})$ is the market wage per unit of effective labor for i 's group, and ϵ_i^w is i 's market ability. Market wages are allowed to differ flexibly across groups and are treated as exogenous in the current model – implying, for example, that marriage or childbearing causally affects market productivity. In practice, however, calibrated group wages may also reflect selection on unobserved ability not captured by the model.²⁵ While the source of wage differences – causal or selection – does not affect the model's first goal of measuring gender norms, it does affect its second goal of counterfactual analysis if selection is likely to change in counterfactual scenarios. Accordingly, in the counterfactual exercises in Section 5, I consider adaptations based on standard assumptions or empirical decompositions from the literature that separate causal effects from selection.

The home production value of individual i is given by

$$h_i = \bar{h}_g^{qr}(\mathcal{K})\epsilon_i^h, \quad g \in \{M, F\}; \quad \text{and} \quad \bar{h}_F^{qr}(\mathcal{K}) = \bar{h}_F^r(\mathcal{K}).$$

$\bar{h}_g^{qr}(\mathcal{K})$ represents the group-specific component of home productivity. For women, I further assume that it is unaffected by marital status or spousal education, conditional on own education and children composition. The assumption allows me to disentangle norm wedge parameters from home productivity parameters. Multiple strands of evidence from time use analyses and home production function estimation suggest that this assumption might be reasonable.²⁶ Nonetheless, I explore the implications of relaxing this assumption in an extension of the model (Appendix D.4), in which non-labor income – particularly husband's earnings for married women – influence the relative productivity of home production versus market work.²⁷

²⁴A sizable share of singles have children in the data, because singles include never-married, divorced, separated, and widowed individuals. This grouping of singles is equivalent to assuming that divorces, separations, and widowhoods occur via shocks exogenous to the schooling years of the couple.

²⁵For example, the extensive literature on returns to schooling pioneered by [Griliches \(1977\)](#) highlights the correlation between schooling and unobserved abilities.

²⁶First, [Ramey and Ramey \(2010\)](#) estimate a minuscule coefficient on household income in explaining mothers' childcare time once their own education and the number and ages of children are controlled for – even among mothers who do not work for pay – suggesting that income effects stemming from a husband's earnings are likely to be small. They use the U.S. Bureau of Labor Statistics time-use surveys, 2003–2008. Second, home production time of married couples in American Time Use Surveys 2003–2006 have been found not to depend on spousal relative wages ([Connelly and Kimmel, 2009](#)) or relative education ([Gupta and Stratton, 2008](#)). Third, [Goussé et al. \(2017\)](#) report no significant differences in home production function parameters by marital status for women, using the British Household Panel Survey, 1991–2008. They also find that even own education has little effect on home production specialization parameters within couples, suggesting spousal education, too, is unlikely to be important.

²⁷This extension requires another assumption: that married men always work.

I assume that market abilities ϵ^w and home production abilities ϵ^h are drawn independently and identically from the Fréchet distribution with shape parameter θ , after the marriage matching stage. The Fréchet distribution has been shown to have a similarly good fit with earnings as more commonly used distributions such as the lognormal (Pindado et al., 2017). The cumulative distribution functions for these abilities are $F(\epsilon^w) = F(\epsilon^h) = F(x) = \exp \{-x^{-\theta}\}$.

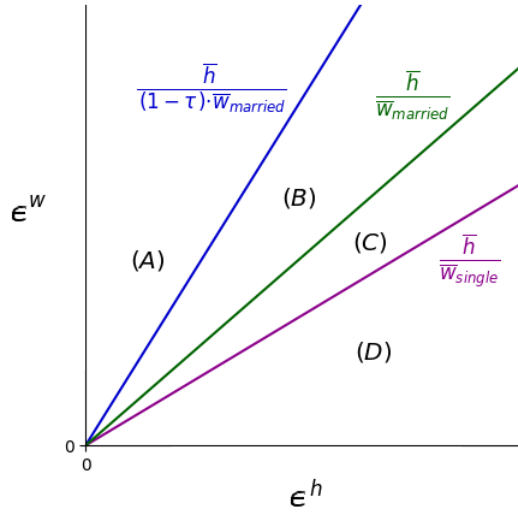
From the property of Fréchet distributions, the probability that a woman in a (q, r) match with children composition $\mathcal{K}$ works in the labor market is given by:

$$P_F^{qr}(\mathcal{K}) \equiv \mathbb{P}((1 - \tau^{qr}(\mathcal{K}))\bar{w}_F^{qr}(\mathcal{K})\epsilon^w > \bar{h}_F^r(\mathcal{K})\epsilon^h) = \frac{[(1 - \tau^{qr}(\mathcal{K}))\bar{w}_F^{qr}(\mathcal{K})]^\theta}{[(1 - \tau^{qr}(\mathcal{K}))\bar{w}_F^{qr}(\mathcal{K})]^\theta + [\bar{h}_F^r(\mathcal{K})]^\theta} \quad (10)$$

The maximum likelihood estimator of this probability is the labor force participation rate of the women in this group. Equation (10) is useful for calibrating parameters in Section 4. Another useful implication of Fréchet abilities is that the average wage of women working in the market is:

$$\begin{aligned} \text{avgwage}_F^{qr}(\mathcal{K}) &= \bar{w}_F^{qr}(\mathcal{K}) \mathbb{E}[\epsilon^w | (1 - \tau^{qr}(\mathcal{K}))\bar{w}_F^{qr}(\mathcal{K})\epsilon^w > \bar{h}_F^r(\mathcal{K})\epsilon^h] \\ &= \bar{w}_F^{qr}(\mathcal{K}) \left(\frac{1}{P_F^{qr}(\mathcal{K})} \right)^{\frac{1}{\theta}} \Gamma\left(1 - \frac{1}{\theta}\right) \end{aligned} \quad (11)$$

Figure 4: Sorting Across Market Work and Home Production of Married and Single Women



Note: This figure shows how labor is allocated between market work and home production for different combinations of market abilities ϵ^w and home abilities ϵ^h , for married and single women with the same education level and children composition (in simplified notation).

- (A): For a married woman to work in the market, her relative talent in market work must be very high.
- (B): If the norms wedge τ was removed, more married women would engage in market work.
- (C): Along with (A) and (B), (C) covers the ability combinations of single women working in the market.
- (D): Single women who work at home have very high relative talent in home production.

Additionally, Figure 4 illustrates how sorting across market work and home production by market and home abilities occurs for married and single women.

3.2 Marriage market

Marriage matching occurs based on the expected value of the economic and noneconomic utilities of each match.

Following Choo and Siow (2006)'s matching under transferable utility (TU) with random preferences, consider an economy consisting of S types of men and women. These types are defined by the level of education determined prior to the matching stage. Denote n^{qr} as the number of marriages between type- q men and type- r women, n^{q0} as the number of single type- q men, and n^{0r} as the number of single type- r women. Also, M^q is the number of type- q men, and F^r the number of type- r women. The following accounting identities must hold:

$$n^{q0} + \sum_{r=1}^S n^{qr} = M^q \quad \forall q = 1, \dots, S \quad (12)$$

$$n^{0r} + \sum_{q=1}^S n^{qr} = F^r \quad \forall r = 1, \dots, S \quad (13)$$

In this TU model, a type- q man must transfer π^{qr} amount of utility to a type- r woman to marry her. The utility of type- q man m marrying a type- r woman at time t is

$$V_m^{qr} = \mathbb{E}(v_m^{qr}) - \pi^{qr} + \psi^{qr} + \varepsilon_m^{qr} \quad (14)$$

where $\mathbb{E}(v_m^{qr})$ is the expected economic utility of man m married to a type- r woman, ψ^{qr} is the noneconomic utility ("marital bliss") enjoyed by the couple, and ε_m^{qr} is m 's random preference for the match drawn independently and identically from the type I extreme-value distribution. Equation (14) encompasses the case of singlehood, denoted by $r = 0$, with $v_m^{q0} \equiv \hat{v}_m^q$, $\pi^{q0} = 0$, and $\psi^{q0} = 0$.

Similarly, the utility of type- r woman f marrying a type- q man is

$$V_f^{qr} = \mathbb{E}(v_f^{qr}) + \pi^{qr} + \psi^{qr} + \varepsilon_f^{qr}$$

where ε_f^{qr} is f 's random preference for the match drawn independently and identically from the type I extreme-value distribution.

The marriage market-clearing condition²⁸ gives an equation for the number of matches:

$$\frac{n^{qr}}{\sqrt{n^{q0}n^{0r}}} = \frac{\mathbb{E}(\hat{v}^{qr}) - \mathbb{E}(\hat{v}_m^q) - \mathbb{E}(\hat{v}_f^r)}{2} + \Psi^{qr} \quad (15)$$

²⁸See Appendix D.6 for more details on the derivation of the marriage market equilibrium.

where $\Psi^{qr} \equiv \psi^{qr} + \frac{\ln \mu^{qr} - 2 \ln(1 + \mu^{qr})}{2}$. The first term on the right-hand side of equation (15) captures the economic gain from marriage relative to singlehood, from the couple enjoying a greater consumption of the public good together. Ψ^{qr} signifies the utility from marital bliss and from the intra-household allocation of resources based on Pareto weights.²⁹

3.3 Education

In this section, I describe the women's educational choice problem, without loss of generality. Woman f chooses the education level with the maximum expected utility:

$$\max_{r=1,\dots,S} \{U_F^r - \xi^r\}$$

where

$$U_F^r = \sum_{q=0}^S \left[\frac{n^{qr}}{F^r} \left(\mathbb{E}(v_f^{qr}) + \pi^{qr} + \psi^{qr} \right) \right] - c_F^r \quad (16)$$

Individuals are forward-looking. The expected utility from schooling level r depends on the consequent matching probabilities in the marriage market and the expected utility in each type of match. The costs of schooling level r , on the other hand, consist of the gender-specific direct utility cost c_F^r and idiosyncratic cost ξ^r , drawn independently and identically from the Type I extreme value distribution.

The distribution of the idiosyncratic schooling costs imply that the probability an individual of gender g chooses schooling level s is

$$\mathbb{P} \left(s = \arg \max_{s'=1,\dots,S} \{U_g^{s'} - \xi^{s'}\} \right) = \frac{\exp\{U_g^s\}}{\sum_{s'=1}^S \exp\{U_g^{s'}\}}$$

The maximum likelihood estimator of this probability is $\frac{F^s}{\sum_{s'=1}^S F^{s'}}$ for women and $\frac{M^s}{\sum_{s'=1}^S M^{s'}}$ for men, i.e. the shares of individuals with education level s for each gender.

3.4 Firms

A representative firm produces market output Y^{mkt} . Although the model includes two market goods – the private good and the public good – I assume both are derived from the same market output. The relative price p simply reflects how much more market output is required to produce one unit of the public good compared to the private good. This simplification is innocuous, as the value of p has no impact on equilibrium outcomes related to education, marriage, or labor supply decisions.

²⁹ ψ_{qr} and μ_{qr} are not separately identified. As shown in Appendix F.3, identifying μ_{qr} would require observing equilibrium transfers in the marriage market, which are not available in the data. I therefore focus on identifying Ψ_{qr} , which is sufficient for conducting counterfactual analyses.

In the baseline model, I adopt the simplest setup on the firm's side, assuming a linear production function in male and female effective labor, $\mathcal{M}$ and $\mathcal{F}$:

$$Y^{mkt} = B(\mathcal{M} + \mathcal{F}) \quad (17)$$

The linear production function implies that market wages are independent of aggregate labor supply. I examine the implications of relaxing this assumption in the counterfactual analysis in Section 5 – when $\mathcal{M}$ and $\mathcal{F}$ may change – by also considering a CES production function in male and female labor, adopted from [Acemoglu et al. \(2004\)](#):

$$Y^{mkt} = B[(1 - \lambda)\mathcal{M}^r + \lambda\mathcal{F}^r]^{\frac{1}{r}} \quad (18)$$

Returning to the baseline linear production function, normalize the labor provided by single males with schooling level 1, market ability 1, and zero children ($\mathcal{K}_0$), i.e. $\bar{w}_M^{10}(\mathcal{K}_0)$, as one unit of effective labor. Then,

$$\begin{aligned} \mathcal{M} &= \sum_{q=1}^S \sum_{r=0}^S \sum_{\mathcal{K}} n^{qr} d^{qr}(\mathcal{K}) \left(\frac{\bar{w}_M^{qr}(\mathcal{K})}{\bar{w}_M^{10}(\mathcal{K}_0)} \right) \left(P_M^{qr}(\mathcal{K}) \right)^{1-\frac{1}{\theta}} \Gamma \left(1 - \frac{1}{\theta} \right) \\ \mathcal{F} &= \sum_{r=1}^S \sum_{q=0}^S \sum_{\mathcal{K}} n^{qr} d^{qr}(\mathcal{K}) \left(\frac{\bar{w}_F^{qr}(\mathcal{K})}{\bar{w}_M^{10}(\mathcal{K}_0)} \right) \left(P_F^{qr}(\mathcal{K}) \right)^{1-\frac{1}{\theta}} \Gamma \left(1 - \frac{1}{\theta} \right) \end{aligned}$$

3.5 Aggregate output

Aggregate output is the sum of market output and home production:

$$Y = Y^{mkt} + Y^{home}$$

where

$$Y^{home} = B \sum_{g \in \{M, F\}} \sum_{(q, r)} \sum_{\mathcal{K}} n^{qr} d^{qr}(\mathcal{K}) \left(\frac{\bar{h}_g^r(\mathcal{K})}{\bar{w}_M^{10}(\mathcal{K}_0)} \right) \left(1 - P_g^{qr}(\mathcal{K}) \right)^{1-\frac{1}{\theta}} \Gamma \left(1 - \frac{1}{\theta} \right) \quad (19)$$

3.6 Equilibrium

An equilibrium in this economy consists of schooling choice q for a man, schooling choice r for a woman, marital transfers π^{qr} , marriage matches (q, r) , public consumption Q , private consumption C_i , labor market participation L_i , total efficient male labor $\mathcal{M}$, total efficient female labor $\mathcal{F}$, market wage, market output Y^{mkt} , total home production Y^{home} , and aggregate output Y , such that

1. Individuals choose the education level that maximizes expected utility, taking as given the probability of different matches and the associated expected utilities.

2. After schooling choices are made, equilibrium marital transfers $\{\pi^{qr}\}$ equate the supply and demand for each marriage match (q, r) .
3. After the matching stage and exogenous determination of children composition, each individual chooses public good consumption Q , private good consumption C_i , and labor supply L_i to maximize their utility. Married individuals maximize equation (1) jointly with their spouse, while singles maximize equation (7) independently.
4. A representative firm hires effective male labor $\mathcal{M}$ and effective female labor $\mathcal{F}$, and pays wage equal to the technology parameter B in equation (17).³⁰
5. Market output Y^{mkt} is given by equation (17), and total home production Y^{home} by (19). Aggregate output Y is given by the sum of Y^{mkt} and Y^{home} .

3.7 Intuition for aggregate output effects of norms wedges

In the model, how does aggregate output respond to changes in the norms wedge? When the norms wedge on married women’s market wages increases, aggregate output can be affected through each of the three stages (in reverse order) of labor supply, marriage, and education.

First, at the labor supply stage, married women’s sorting between market work and home production is less aligned with their comparative advantage. This lowers aggregate output.

Second, at the marriage matching stage, higher disutility from wives working in the market makes marriage less attractive, leading some women who would have married to remain single. These newly single women no longer face the norms wedge, which fosters them to pursue their comparative advantage, thereby increasing output. Another effect is that newly single men and women now receive singles’ wages and home productivities. Wage differences between married and single individuals reflect productivity differences³¹ arising from occupational choices, firm sorting, job mobility, hours, work schedule flexibility, and work effort – factors shaped directly by marital status and indirectly by differential likelihood of having children.³² Whether married men and women are more or less productive in the market and at home than their single counterparts, and how these differences aggregate, is an empirical question. Hence, the net effect on output at the marriage matching stage is a priori ambiguous.

³⁰See Appendix H.2.1 for details on how I solve for the equilibrium under the CES production function (equation (18)) through fixed-point iteration.

³¹These wage differences may also partly reflect selection or discrimination. As a conservative robustness check, I test the sensitivity of my counterfactual results to assuming that marriage has no causal effect on productivity (Figure 11, case 4) by fully suppressing wage differences between married and single individuals, conditional on gender, education, and children composition.

³²In a lifecycle analysis of U.S. wages, Goldin et al. (2024) find that mothers face a “motherhood penalty,” trading lower pay for employment at more family-friendly firms. Fathers, by contrast, receive a “fatherhood premium,” especially in time-intensive jobs, where marriage and parenthood may have enabled or motivated higher work effort.

Third, at the educational choice stage, if there is positive assortative matching on education then the lower likelihood of marriage decreases the young women’s marriage-market incentive to invest in education. In addition, if education raises market productivity more than home productivity, the labor-market incentive to get educated depends on how the expected likelihood of working changes. The likelihood of working in the market depends on the balance between i) the lower likelihood of being subject to the norms wedges as marriage rates fall, and ii) the higher value of the norms wedges once married. How the labor-market returns to education changes is an empirical question, but if education does decrease on net, then aggregate output would also decrease through lower market wages and home productivities.

In sum, the overall effect on aggregate output depends on the relative strengths of these mechanisms and the underlying parameter values.

4 Data and Parameter Inference

4.1 Data

To simulate the U.S. economy within the model framework, I use the U.S. decennial census (Ruggles et al., 2024), consisting of 1-in-100 nationally representative random sample of individuals. An advantage of the census is that it collects information on all household members so that labor market participation is observed for both spouses in married couples. As the presence of other income-earning household members may influence individual labor decisions, I restrict the sample to household heads and their spouses. I further limit the sample to individuals aged 25 to 54, after education is complete and when individuals are the most economically active.³³

The model is fitted to census data for each decade, assuming that the data reflects the model equilibrium. Calibrating the model separately by decade allows all model parameters to vary flexibly over time. These are: children composition probabilities $\{d^{qr}(\mathcal{K})\}$, the inverse measure of the dispersion of market and home abilities $\{\theta\}$, group market wages $\{\bar{w}_M^{qr}(\mathcal{K}), \bar{w}_F^{qr}(\mathcal{K})\}$ and home productivities $\{\bar{h}_M^{qr}(\mathcal{K}), \bar{h}_F^r(\mathcal{K})\}$, gender norms wedges $\{\tau^{qr}(\mathcal{K})\}$, the expected utility enjoyed by a (q, r) match $\{\Psi^{qr}\}$, and the cost of each schooling level $\{c_M^q, c_F^r\}$.

The practical advantage of my model setup is that it is not demanding on the data; the variables required for parameter inference are market wage, labor force participation status, marital status, education, and children. Since all are first jointly available in 1940, I match the model to the decennial census from 1940 to 2010. I exclude the 1950 census, however, because it lacks information on spousal wages. See Appendix E for details on the specific census samples used each year and

³³Appendix Figure A2 shows that the age range of 25-54 is the most economically active 30-year window.

on the calculation of market wages.³⁴

Section 4.2 below transparently describes the parameter inference procedure, which I implement in sequential steps: starting with an estimate of θ , each subsequent parameter is estimated using the results from earlier steps.

Some model parameters, such as group market wages or home productivities, are specific to a group, defined by gender, marriage match (i.e., schooling pair), and children composition. As I estimate them by matching population moments to their sample analogs within each group,³⁵ the groups must be sufficiently large for the law of large numbers to apply. At the same time, because individuals within a group share the same parameters, the grouping must also be sufficiently specific.³⁶ Among the grouping variables, schooling underwent the most substantial changes over the sample period. To account for shifting educational distributions over time, I construct five or six schooling categories for each decade, ensuring that at least 5% of the sample falls into each category. For children composition, I similarly define categories to balance group size with group specificity. Since the most pronounced differences in home production duties relate to the presence of a first child and whether the child is young, these factors form the basis for categorization.³⁷

4.2 Steps for parameter inference

1. $d^{qr}(\mathcal{K})$: probability of a (q, r) match having children composition $\mathcal{K}$

Set at the empirical probabilities.

2. θ : inverse measure of dispersion of market and home abilities

Making use of the fact that wages of individuals working in the market follow a Fréchet distribution, I estimate θ through maximum likelihood. Where x_n is the market ability of observation n and P_n denotes the fraction of workers in observation n 's group, the maximum likelihood estimator for θ is:³⁸

$$\tilde{\theta}_{MLE} = \arg \max_{\theta \in (0, \infty)} \sum_{n=1}^{Nobs} [\ln \theta - \ln P_n - x_n^{-\theta} P_n^{-1} - (\theta + 1) \ln x_n]$$

³⁴Market wages are computed by dividing total pre-tax wage and salary income from the previous year by the total number of hours worked in that year.

³⁵For example, see equations (10) and (11), applied in steps 3 and 4 of Section 4.2.

³⁶Appendix Table A1 shows that education is by far the strongest predictor of attitudes toward the role of married women, followed by survey year and the number of children – which are all incorporated into the group definitions and decade-by-decade calibration.

³⁷The schooling level and children composition categorizations are in Appendix Tables A2 and A3.

³⁸How I derive the likelihood function and how I isolate market abilities from observed market wages are detailed in Appendixes F.1 and F.2, respectively.

3. $\bar{w}_g^{qr}(\mathcal{K})$: group market productivity per unit of effective labor

Using the estimate of θ found in step 2 and the average wage and proportion of market-workers in each group in the data, I can back out $\bar{w}_g^{qr}(\mathcal{K})$ from equation (11).

$$\bar{w}_g^{qr}(\mathcal{K}) = \text{avgwage}_g^{qr}(\mathcal{K}) (P_g^{qr}(\mathcal{K}))^{\frac{1}{\theta}} \frac{1}{\Gamma(1 - 1/\theta)}$$

4. $\bar{h}_M^{qr}(\mathcal{K})$: group home productivity per unit of effective labor

$\bar{h}_M^{qr}(\mathcal{K})$ can be backed out from equation (10), armed with θ found in step 2 and the average wage and proportion of market-workers in each group of men in the data. For $\bar{h}_F^r(\mathcal{K})$, I use θ found in step 2 and the average wage and proportion of market-workers among *single* women with r years of schooling and children composition $\mathcal{K}$.

$$\bar{h}_F^r(\mathcal{K}) = \text{avgwage}_F^{0r}(\mathcal{K}) (1 - P_F^{0r}(\mathcal{K}))^{\frac{1}{\theta}} \frac{1}{\Gamma(1 - 1/\theta)}$$

5. $\tau^{qr}(\mathcal{K})$: group norms wedges

$\tau^{qr}(\mathcal{K})$ is backed out from equation (10) using $\bar{w}_F^{qr}(\mathcal{K})$, $\bar{h}_F^r(\mathcal{K})$, and the fraction of market workers in a group of married women. The idea is that norms wedges are high if the fraction working in the market is much lower than is predicted from market and home productivities.

$$\tau^{qr}(\mathcal{K}) = 1 - \frac{\text{avgwage}_F^{0r}(\mathcal{K})}{\text{avgwage}_F^{qr}(\mathcal{K})} \left(\frac{1 - P_F^{0r}(\mathcal{K})}{1 - P_F^{qr}(\mathcal{K})} \right)^{\frac{1}{\theta}} \quad (20)$$

Intuitively, the disutility from wives working is inferred by comparing the labor choices of married and single women sharing the same level of education and the same children composition. The difference in their labor market participation rates that cannot be explained by wage differentials is attributed to gender norms.

6. Ψ^{qr} : utility from marital bliss and intra-household resource allocation, in a (q, r) match

From equation (15),

$$\Psi^{qr} = \frac{n^{qr}}{\sqrt{n^{q0}n^{0r}}} - 2A^{qr} + \hat{A}_M^q + \hat{A}_F^r. \quad (21)$$

where

$$A^{qr} = \sum_{\mathcal{K}} d^{qr}(\mathcal{K}) \mathbb{E} \left[\ln \left(\bar{w}_M^{qr}(\mathcal{K}) \epsilon_m^w L_m^* + \bar{h}_M^{qr}(\mathcal{K}) \epsilon_m^h (1 - L_m^*) + \right. \right. \\ \left. \left. [1 - \tau^{qr}(\mathcal{K})] \bar{w}_F^{qr}(\mathcal{K}) \epsilon_f^w L_f^* + \bar{h}_F^r(\mathcal{K}) \epsilon_f^h (1 - L_f^*) \right) \right]$$

and

$$\hat{A}_g^s = \sum_{\mathcal{K}} d^{qr}(\mathcal{K}) \mathbb{E} \left[\ln \left(\bar{w}_g^{qr}(\mathcal{K}) \epsilon_i^w \hat{L}_i^* + \bar{h}_g^s(\mathcal{K}) \epsilon_i^h (1 - \hat{L}_i^*) \right) \right] \quad (q = 0, r = s \text{ if } g = F, \text{ and } q = s, r = 0 \text{ if } g = M)$$

There are no closed-form expressions for A^{qr} or $\hat{A}_g^s$ so I simulate them to back out Ψ^{qr} .³⁹

7. c_g^s : utility cost of acquiring education s

Use equation (16). $\{c_F^r\}_{r=1,\dots,S}$ are found as the solution to the system of equations

$$\frac{F^r}{\sum_{r'=1}^S F^{r'}} = \frac{\exp \left\{ \sum_{q=0}^S \frac{n^{qr}}{F^r} H^{qr} - c_F^r \right\}}{\sum_{r'=1}^S \exp \left\{ \sum_{q=0}^S \frac{n^{qr'}}{F^{r'}} H^{qr'} - c_F^{r'} \right\}} \quad \forall r = 1, \dots, S \quad (22)$$

where

$$H^{qr} = \begin{cases} 2\hat{A}_F^r & \text{if } q = 0 \\ 2A^{qr} + \Psi^{qr} + \frac{\ln n^{q0} - \ln n^{0r}}{2} - \hat{A}_M^q + \hat{A}_F^r & \text{if } q = 1, \dots, S \end{cases}$$

Additionally, $\{c_M^q\}_{q=1,\dots,S}$ are found similarly.⁴⁰

4.3 Calibration results and discussion

In this section, I discuss the calibrated parameter values and how they align with comparable estimates in the literature, related measures from external data sources, or well-established stylized facts. I place particular focus on the calibrated norms wedges, τ .

θ : Dispersion of market and home abilities Appendix Table A4 reports the maximum-likelihood estimates of θ , which range from 1.55 to 2.08 across years. These values are close to Hsieh et al. (2019)'s estimate of 1.52 for the Fréchet shape parameter governing the dispersion of abilities across occupations, as well as their chosen value of 2 for counterfactual analysis. In addition, Appendix Figure A3 shows that the inferred distribution of market abilities closely resembles a Fréchet distribution, lending further support to the assumption of Fréchet-distributed abilities.

$\bar{w}, \bar{h}$: Group market and home productivities Group market wage $\bar{w}$ increases with both the average wage of workers in the group and the group's labor force participation (LFP) rate. The dependence on the LFP rate reflects selection effects. Higher LFP implies weaker positive selection

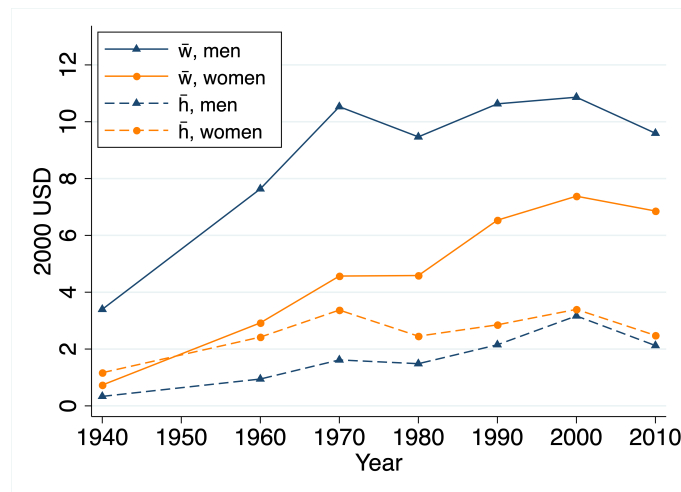
³⁹To permit a more realistic curvature of the utility function, I use the CRRA utility function instead of log utility in simulating A^{qr} and $\hat{A}_g^s$. Note that the CRRA utility fully conforms with the set of conditions of the general utility function, described in Appendix D.1. The coefficient of relative risk aversion used is 0.71, which Chetty (2006) finds to be the average value across 33 different estimation methods spanning structural lifecycle models, natural experiments, earned income responses, and macroeconomic trends.

⁴⁰Detailed derivations are in Appendix F.3.

on idiosyncratic market ability, prompting the model to assign a higher value to $\bar{w}$ to reconcile the observed average wage. In contrast, group home productivity $\bar{h}$ also rises with average wage of the workers in that group but is decreasing in the LFP rate of that group.

Figure 5 plots the averages of $\bar{w}$ and $\bar{h}$ by sex and decade, weighted by the empirical probability of each group. Male market productivity rises steadily from 1940 to 1970 and then plateaus through 2010. Female market productivity, while consistently lower than men's, shows a more sustained and pronounced increase, gradually narrowing the gender wage gap. With the exception of women in 1940, home productivity remains lower than market productivity, though it also grows over time. Women's home productivity is consistently higher than men's, but the gap narrows over time. These patterns are consistent with rising female attachment to the labor market and persistent – but gradually weakening – gender differences in comparative advantage across market work and home production.

Figure 5: Group Market Productivity ($\bar{w}$) and Home Productivity ($\bar{h}$) by Sex



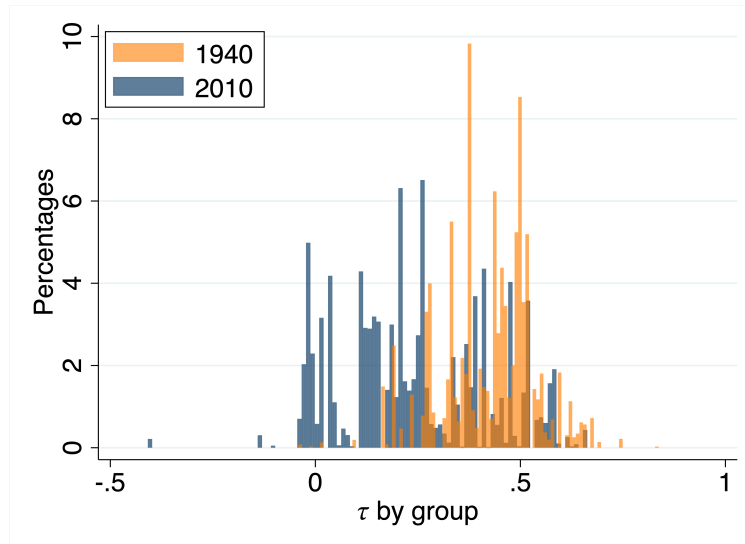
Note: This figure plots the weighted average of the group components of market productivity and home productivity by sex and decade.

How do $\bar{w}$ and $\bar{h}$ vary by education? As discussed in Section 3.7 regarding the intuition behind aggregate productivity effects from decreases in norms wedges, the relative increase of $\bar{w}$ versus $\bar{h}$ with education is crucial for educational choices in counterfactual scenarios. Specifically, if a young woman anticipates a higher likelihood of market work due to a reduction in norms wedges, she will choose to invest more in education only if education raises market productivity more than home productivity. Appendix Table A5 confirms that, for both sexes, education increases market productivity more than home productivity.

τ : Gender norms wedges τ represents the norms wedge on the market wage of married women. To illustrate, a value of $\tau = 0.6$ means that a married woman perceives a \$10 market wage as effectively worth only \$4 when making her labor supply decision.

What exactly does τ encompass? Though labeled as “norms wedges”, τ is a catch-all term that includes any reason why married women’s LFP differs from that of single women beyond wage differences. It is important to be clear about what factors are – and are not – captured by τ . Because the effect of wage differentials are removed out, τ is *not* confounded by differences in market abilities between married women and similar single women. On other other hand, τ does include many factors that fall under the broad concept of homemaker norms for married women, either self-imposed or expected by others: a) the preference to conform to traditional gender norms as a wife, b) differential valuation of staying home for married women relative to single women,⁴¹ c) differential non-wage treatment of married women by employers relative to single women, and d) men’s preference to have more home-productive women as wives. All these justify referring to τ as “norms wedges.” One factor included in τ that does not fall under homemaker norms is that married women make labor supply decisions within two-person households where specialization is possible, whereas single women operate within one-person households. I address this issue by also calibrating the extension with income effects (Appendix D.4) that directly incorporates this difference, discussed further below.

Figure 6: Histogram of Norms Wedges on Married Women’s Market Wages (τ)



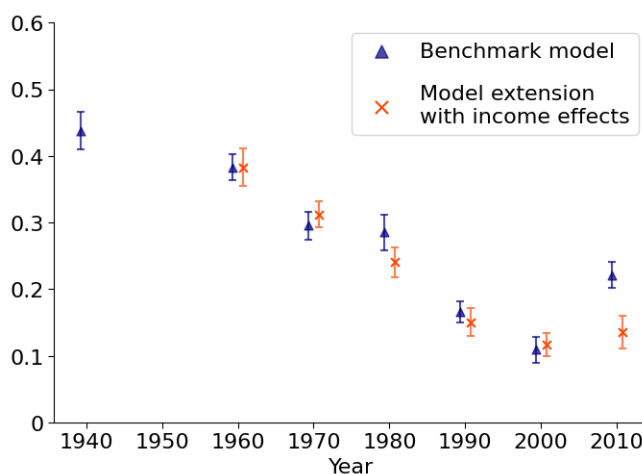
Note: This figure plots the histogram of the calibrated $\tau^{qr}(\mathcal{K})$ values for 1940 and 2010. Each group is weighted by its empirical probability.

⁴¹See Appendix D.3 for an explicit demonstration.

Figure 6 plots the weighted histograms of τ , calibrated by group, for the years 1940 and 2010. The 2010 distribution clearly sits to the left of the 1940 distribution, indicating a decline in norms wedges over time.

I take the inferred group-specific τ as noisy estimates of norms wedges, as there are many selection effects unaccounted for in the model, such as the correlation between education choice and market abilities, or the correlation between taste for spousal type and market abilities. Therefore, I consider the weighted median of τ by decade.⁴² Figure 7 plots these values in blue. τ generally decreases over time, except from 2000 to 2010. The mathematical explanation for the rise at the end is that, although married women's wages increased relative to those of single women during this period, their LFP did not.⁴³ To reconcile these observations, τ must increase.

Figure 7: Evolution of Norms Wedges τ



Note: This figure plots the weighted median of τ by decade, in the benchmark model and in the model extension with income effects (Appendix D.4). The weight of each group corresponds to its empirical probability in the data. The error bars indicate 95% confidence intervals based on bootstrapped standard errors from 100 replications. The τ in the extension can only be estimated from 1960, because household income beyond one's own earnings (i.e., spousal earnings, welfare benefits, social security income, unearned income) are reported only from 1960.

Further insight comes from comparing the τ values in the benchmark model to those in the extension with income effects, as shown in Figure 7. In the extension, married women's labor supply decisions respond to their husbands' market earnings; household income effects reduce their incentive to contribute additional market income. Notably, τ barely rises between 2000 and 2010

⁴²The results are very similar when using the weighted average, unweighted average, or unweighted median. I prefer the median, as it is less sensitive to outliers that arise in groups with low empirical probability.

⁴³Figure 1 illustrates that female LFP has plateaued since 1990. The rise in τ persists even when considering market work *hours* of married women relative to single women, indicating that it is not driven by the simple LFP measure failing to capture increases in hours.

in the extension, indicating it is the failure to capture spousal dependence that leads the benchmark model to overestimate τ in 2010. Nonetheless, the benchmark model performs well up to 2000.

Table 1: Correlation between Norms Wedges and Attitudinal Survey Answers

	State-level attitudes		State $\times$ decade-level attitudes	Individual-level attitudes	
	Against married women working	All questions	Against married women working	All questions	
	(1)	(2)	(3)	(4)	(5)
τ by state	0.487*** (0.149)	0.239*** (0.085)			
τ by state $\times$ decade			0.237** (0.110)	0.049* (0.028)	0.049* (0.029)
N	51	51	233	188,333	134,661
State FE	-	-	Yes	Yes	Yes
Decade FE	-	-	Yes	Yes	Yes
Question FE	-	-	-	Yes	Yes
Demographic controls	-	-	-	No	Yes

Note: This figure shows the correlation between model-based norms wedges and survey measures of attitudes. The attitudinal questions (from Figure 3) are coded so that higher values indicate more traditional views. State-level and state $\times$ decade-level attitudes are computed as sampling-weighted averages of individual responses. “Against married women working” corresponds to the question: *Should a married woman earn money if she has a husband capable of supporting her?*, asked 1938-1998. Columns 1-2 show cross-sectional regressions at the state level, weighted by state population (robust standard errors). Column 3 reports a state $\times$ decade-level regression with state and decade fixed effects, weighted by population (standard errors clustered by state). I do not estimate a state $\times$ decade-level regression using all questions because the questions vary in timing, making attitudes aggregated at that level non-comparable over time. Instead, columns 4-5 regress individual responses to the 14 questions on state $\times$ year-level τ , with state, decade, and question fixed effects (standard errors clustered by state). Column 5 adds demographic controls (age, sex, and race dummies) for greater precision, but some questions then drop out due to missing demographic information. * $p < 0.10$, ** $p < 0.05$ *** $p < 0.01$

The norms wedges, though simple to impute, meaningfully capture gender norms in marriage. Table 1 shows that the values of norms wedges strongly correlate with survey-based measures of attitudes about gender roles in marriage. I redo the parameter inference procedure to compute τ at both the state and state $\times$ decade levels.⁴⁴ For attitudes, I use either all 14 attitudinal questions on gender roles within marriage in Figure 3, or a single question that has been asked the most

⁴⁴To obtain geographic variation in τ , I redo the parameter inference procedure. The group categories are reformulated to ensure, as before, that each group is specific enough but also large enough. For τ by state (state $\times$ decade), a group is defined by state, schooling pair, age cohorts – ages 25-34, 35-44, 45-54(, and decade). Then I compute for each state (state $\times$ decade) the weighted median of the group-level τ .

consistently over time (1938-1998): whether a married woman should work.⁴⁵ States with higher τ are also those where a greater share of respondents disapprove of married women working (column 1)⁴⁶ and answer more conservatively on the 14 questions combined (column 2). The relationship holds also *within state* over time. τ at the state $\times$ decade level remains strongly correlated with the state $\times$ decade average response to the single question, even with state fixed effects (column 3). To incorporate all 14 questions, I also run an individual-level regression of the survey answers on τ at the state $\times$ decade level, with question fixed effects and state fixed effects. The positive relationship persists, albeit more weakly, both with and without individual-level demographic controls (columns 4-5).

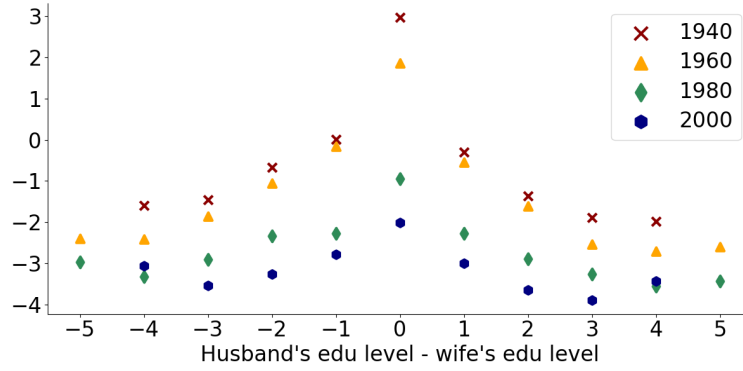
Finally, Appendix G.2 demonstrates the robustness of τ to sample perturbations and alternative assumptions, including several that address additional differences between married and single women unrelated to cultural norms. These include: 1) incorporating differential tax treatment by marital status by using imputed after-tax wages instead of gross wages; 2) including business and farm income in addition to salary income (from 1960); 3) holding θ fixed, as done in Hsieh et al. (2019), instead of allowing it to vary by decade; 4) dividing married and single individuals by legal marital status instead of using the IPUMS-constructed “probable spouse” measure, which includes cohabiting couples from 1990; 5) excluding widows from single women to assess sensitivity to the composition of singles; 6) limiting the sample to individuals aged 40–54, rather than 25–54, to account for differences in childbearing potential between married and single women; 7) fixing education levels in all decades (less than high school, high school graduate, more than high school).

Ψ : Utility from marital bliss and intra-household resource allocation Figure 8 plots the average Ψ for each value of the spousal education gap by decade. Within each decade, Ψ exhibits an approximately single-peaked shape, with the peak occurring when spouses have the same education level. Moreover, the decline in the values of Ψ over time indicates a reduction in the overall attractiveness of marriage. These patterns align with the well-documented prevalence of assortative matching by education and falling marriage rates in the U.S. (Greenwood et al., 2016).

⁴⁵The single question actually combines responses from two nearly identical questions – [1] and [2] in Appendix Figure A1. See the figure for the exact wordings. Question [1] was asked in the Gallup Polls, and question [2] in the General Social Survey. As shown in the figure, the yearly average trends of the two questions closely overlap, lending confidence that they can be treated as a single question.

⁴⁶Appendix Figure A4 visually shows the tight relationship in column 1.

Figure 8: Ψ by Spousal Education Gap



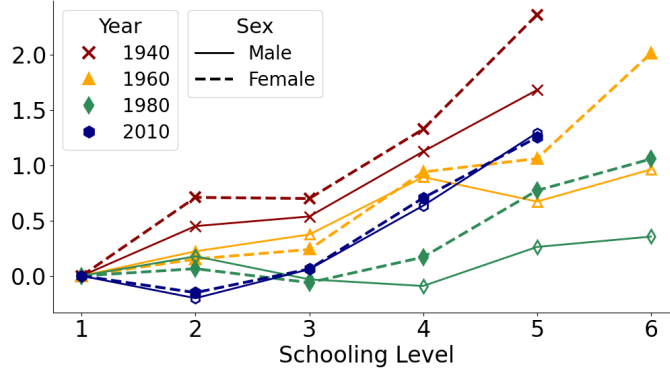
Note: This figure plots, by decade, the average Ψ^{qr} for each value of the difference in the husband's and the wife's education levels ($q - r$), weighted by the empirical probability of each marriage match (q, r).

c: Utility cost of schooling The final set of parameters is the gender-specific cost of schooling, c_g^s . These costs are inferred by comparing observed educational attainment to the levels predicted by expected utility, which incorporates both marriage-market returns and economic (wage and home productivity) returns. For example, if more women attain high school than the model predicts based on expected utility, the inferred cost of that level for women must be relatively low.

Figure 9 plots the values of c_g^s in selected years, divided by the average economic utility (A^{qr}) of that year for greater comparability over time.⁴⁷ It shows that the estimated cost of schooling c increases with education level in all years. While changes in education categories across decades limit direct comparisons over time, broad patterns emerge. Initially, the costs decline – consistent with rising educational attainment – but rise again in later decades, potentially reflecting increasing pecuniary costs of post-secondary education. Within each decade, c values are comparable across sex. Early cohorts show higher costs for women, especially at advanced education levels, but the gender gap narrows over time. By 2010, male and female cost profiles are nearly identical, and the male cost of attaining the highest schooling level even exceeds the female cost. This pattern reflects the well-documented reversal in gender gaps in educational attainment in the U.S., where women began surpassing men in college completion rates from the mid-1990s.

⁴⁷Raw values of c_g^s for every decade are reported in Appendix Table A6.

Figure 9: Cost of Schooling c by Sex and Decade



Note: This figure plots, by decade, the utility cost of schooling (c_g^s) normalized by the average economic utility (A^{qr}) for that decade. c_g^1 is normalized to zero for each year and gender.

5 Counterfactual Exercises

To quantify the contribution to economic growth of changes in gender roles in marriage, I will consider how aggregate output Y responds to changes in the norms wedges, while holding all other parameters fixed. The steps for conducting counterfactuals are described in Appendix H.1. Moreover, I benchmark the results from this main counterfactual against the effects of other counterfactuals described below.

Main Counterfactual Table 2 shows how various aggregate variables in 2000 – the latest decade in which the benchmark model performs well – would change if gender norms had remained at their 1940 level, holding all other parameters fixed at 2000 values. Because group-specific norm wedges are likely noisy estimates, I use the weighted median τ for each decade (plotted in Figure 7). Applying the same value of τ to all individuals in the counterfactuals also ensures that changes in group sizes across scenarios do not mechanically alter the mean or distribution of τ in the economy.

I consider two adjustment margins: when only labor supply responds (column 1), and when education, marriage, and labor supply all respond (column 2).

Column 1 isolates the direct effect of gender norms by holding marriage and education choices fixed. When τ rises from 0.11 in 2000 to 0.44 in 1940 – reflecting more traditional norms for married women – their labor force participation drops by 24.8%. This reduces their cumulative market output Y^{mkt} by 12.3%. At the same time, home production Y^{home} increases, offsetting part of the loss. As a result, their cumulative total output $Y = Y^{mkt} + Y^{home}$ declines by only 2.9%. In the aggregate, factoring in the share of married women in the population, market output falls by 3.6% and total output by 0.9%. The dissimilar effects on Y^{mkt} and Y highlight the importance of accounting for nonmarket output, often excluded from national accounts. The output gains when

women enter the labor market would be overstated in methods that only consider market output.

**Table 2: Percentage Changes in Various Aggregate Variables
if Individuals of 2000 were Subject to the Norms Wedge of 1940**

	Adjustment margins: Labor supply (1)	Education, marriage, labor supply (2)
Education		
Women's average years of schooling	-	-2.6
Men's average years of schooling	-	-2.0
Selection into marriage		
Marriage rate	-	-26.3
Married women's average years of schooling	-	-5.3
Single women's average years of schooling	-	1.2
Married men's average years of schooling	-	-4.3
Single men's average years of schooling	-	0.9
Latent market and home productivity		
Women's average $\bar{w}$	-	-2.9
Men's average $\bar{w}$	-	-7.2
Women's average $\bar{h}$	-	-0.4
Men's average $\bar{h}$	-	3.3
Labor Force Participation		
Married women's LFP	-24.8	-30.8
Single women's LFP	-	0.8
Married men's LFP	-	-1.6
Single men's LFP	-	0.4
Output per head		
Married women's market output	-12.3	-23.8
Married women's total output	-2.9	-10.6
Married men's market output	-	-7.6
Married men's total output	-	-6.2
Aggregate market output	-3.6	-8.7
Aggregate total output	-0.9	-5.3

Note: This table reports the percentage changes in various aggregate variables that occur when the individuals of 2000 are subject to the norms wedge of 1940, holding all other parameter values constant at the 2000 level. Column 1 holds the marriage match patterns and educational choices constant at the 2000 level and considers only changes to the married individuals' labor supply decisions. Column 2 additionally allows the (forward-looking) marriage matching and educational choices to change in accordance with the new expected utilities arising from the altered labor supply behavior.

In column 2, higher τ not only affects labor supply directly but also indirectly alters education and marriage decisions. Since norm wedges are modeled as marriage-specific costs, a higher τ

makes marriage less attractive, leading to a 26.3% drop in the marriage rate. As the norms wedge is more costly for women with higher market wages, higher-educated women are the first to drop out of marriage. Due to assortative marriage matching by education, the fall in marriage is more pronounced also for higher educated men. As the set of married women is now more negatively selected on education, married women's LFP rate and cumulative market output fall by more in column 2 than in column 1. Married women's cumulative total output falls by 10.6%.

Furthermore, average years of schooling decline for both men (by -2.0%) and women (-2.6%), with a more pronounced drop among women.⁴⁸ For men, this decline is driven solely by reduced returns in the marriage market. For women, education decisions respond to changes in both marriage- and labor-market returns. The direction of change in labor-market returns is *ex ante* ambiguous. Highly educated women, now more likely to remain single, avoid the norms wedge and are more likely to participate in the labor market. However, among those who marry, traditional norms reduce their likelihood of market work. In this counterfactual, women's labor-market returns to education appear to decline as well, contributing to a larger reduction in educational attainment among women relative to men.

Lower education and marriage rates alter men's and women's latent productivity in the market and at home, captured by the average of $\bar{w}_g^{qr}(\mathcal{K})$ and $\bar{h}_g^{qr}(\mathcal{K})$ by gender. Interestingly, although women's education declines more than men's, their market productivity falls by less (-2.9% for women vs. -7.2% for men). This is because married men have higher $\bar{w}$ than single men, whereas the opposite holds for women.⁴⁹ Thus, for women, lower marriage rates partially offset the productivity loss from lower education; for men, both factors reduce market productivity. Home productivity decreases slightly for men and increases slightly for women, consistent with its modest sensitivity to education (Appendix Table A5) and the fact that single men have higher home productivity than married men.

Overall, aggregate market output falls by 8.7% and total output by 5.3%. Comparing the output counterfactuals in columns 1 and 2 highlights the importance of accounting for general equilibrium effects. The difference is striking: allowing for endogenous responses in education and marriage markets nearly sixfolds the total output loss from 0.9% to 5.3%. The comparison demonstrates that partial equilibrium analyses, as is often the case with regression-based empirical research, may severely underestimate the true economic costs of traditional gender norms.

Is the effect of a 8.7% fall in aggregate market output and a 5.3% fall in aggregate total output small or large? Table 3 shows that these drops in aggregate output represent 12.6% of the growth

⁴⁸The decline is even starker when measured by the percentage change in the share attaining the highest education level: -10.7% for men, and -14.1% for women.

⁴⁹This pattern aligns with Goldin et al. (2024), who document a motherhood penalty (primarily driven by lower work hours and sorting into family-friendly firms with lower pay) and fatherhood premium (potentially driven by higher work effort) in wages in the U.S.

in market output and 8.1% of the growth in total output between 1940 and 2000. The effects might be viewed as large, as the norms wedge parameters capture quite a narrow concept of gender norms relating to the distinction between married and single women who are otherwise similar. On the other hand, it might be viewed as modest relative to the overall scale of output growth that has occurred over 1940-2000. To benchmark the magnitude of these output effects from another angle, I conduct some additional counterfactuals below.

Table 3: Share of Growth Accounted for by Norms Wedge

	Actual 1940 (1)	Actual 2000 (2)	Counterfactual 2000 with 1940 τ (3)	1940-2000 growth accounted for by τ (4)
Aggregate market output (2000 USD)	4.7	15.1	13.8	12.6%
Aggregate total output (2000 USD)	6.2	17.5	16.6	8.1%
Share of women with >12 yrs of schooling (%)	10.0	52.8	45.4	17.4%
Share of men with >12 yrs of schooling (%)	11.7	53.1	47.4	13.7%
Share of married women with >12 yrs of schooling (%)	9.9	52.6	36.3	38.3%
Married women's LFP (%)	14.4	74.6	51.7	38.1%
Wives' share of household market income (%)	6.3	36.3	27.3	30.0%

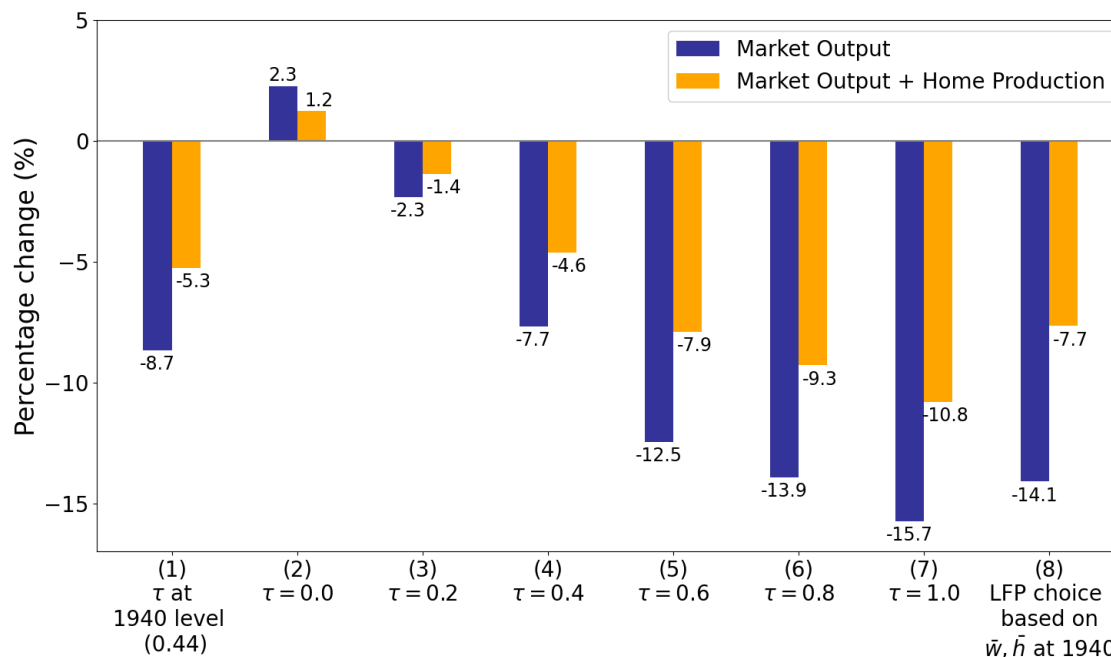
Note: Columns 1-3 report the levels of various aggregate variables in 1940, 2000, and in the counterfactual 2000 economy with 1940 norms. Column 4 shows the share of growth attributed to the change in τ between 1940 and 2000. Aggregate market output is normalized to equal the average real hourly wage, while aggregate total output is normalized to equal the sum of the average real hourly wage and the average real hourly home productivity. Wife's share of household market income is her real hourly wage divided by the couple's total wage, with nonparticipation counted as zero. These hourly measures reflect the model's binary labor supply choice – market work or home production – excluding the role for hours worked.

Besides aggregate productivity growth, the decline in gender norms wedges between 1940 and 2000 helps explain several key changes in U.S. economic and social outcomes over this period. As shown in Table 3, changing norms account for 38.1% of the rise in married women's labor force participation and 30.0% of the increase in wives' share of household market income. The decline in gender norms also contributed to growing educational attainment for both men and women, but since it accounts for a larger share of women's gains (17.4%) than men's (13.7%), it helps explain why women's education grew faster than men's during this period. Moreover, changing norms account for 38.3% of the increase in married women's education – a substantially larger share than for all women's education. Thus, the decline in τ also helps explain the increasingly positive selection into marriage by education.⁵⁰

⁵⁰The increasingly positive selection into marriage by education is not clearly demonstrated in this table. It is typically demonstrated using college vs. non-college distinctions, as in Bar et al. (2018), who document growing

Additional Counterfactuals Figure 10 compares the market and total (market and nonmarket) output effects of various counterfactual scenarios, where the baseline year is set at 2000. Besides the main counterfactual (scenario 1), the others explore the effects of τ changing from 0 to 1 (scenarios 2-7), and when labor supply choices of females are made based on $\bar{w}$ and $\bar{h}$ of 1940 (scenario 8). All the three margins of education, marriage, and labor supply are allowed to adjust.

Figure 10: Market and Total Output Effects of Different Counterfactuals



Note: This figure compares the market and total (market and nonmarket) output effects of various counterfactual scenarios, where the baseline year is set at 2000, with τ at 2000 being 0.15.

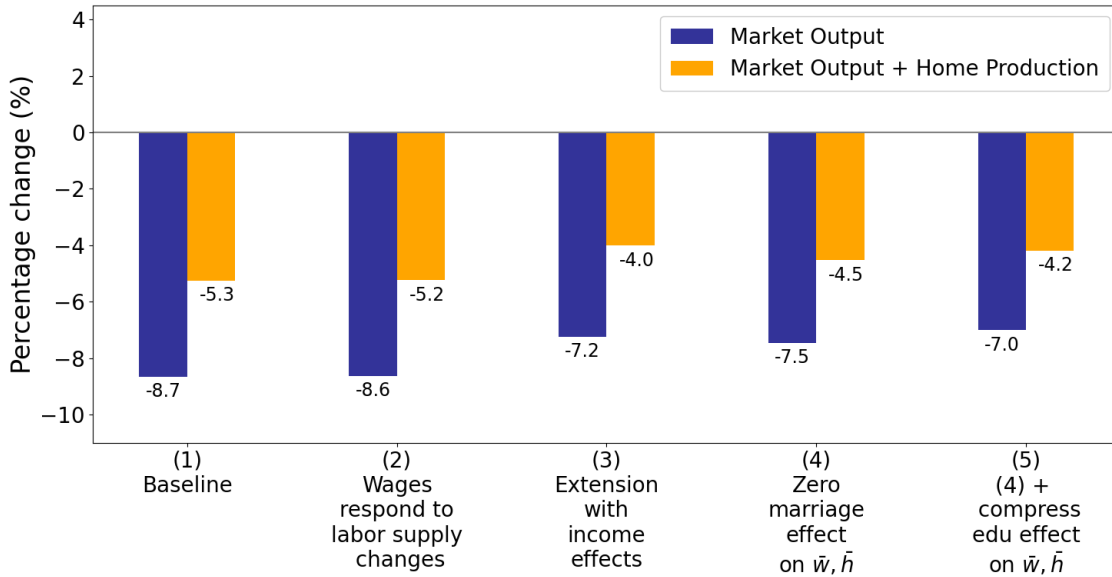
As τ increases, the negative output effects becomes increasingly pronounced. When married women are completely prohibited from working ($\tau=1$), market output falls by 15.7% and total output by 10.8%. This counterfactual highlights the potential loss in aggregate output incurred in countries with the most conservative attitudes towards married women working, or the loss that could result in the U.S. if the views of a small but vocal and increasingly visible segment of religious traditionalists, Christian nationalists (Whitehead and Perry, 2020), and online communities (Ging, 2019) – who advocate a return to traditional gender roles and oppose married women’s participation in the workforce – were to become the societal norm.

marriage rate of women with at least a college degree relative to that of women with less education. I, too, find that married women’s college education share is higher than single women’s in 2000 (28.5% vs. 28.0%). In Table 3, however, I focus on the >12 years of schooling threshold for consistency between 1940 and 2000; the share of women holding college degrees was extremely small in 1940.

I focus on counterfactual scenario 8 to benchmark the effects of the main counterfactual. As illustrated in Figure 5, market and home productivities changed substantially between 1940 and 2000, fueling economic growth over this period. Notably, women's $\bar{w}$ was lower than their $\bar{h}$ in 1940, whereas by 2000, $\bar{w}$ was more than twice as large as $\bar{h}$. To avoid capturing the direct effects of productivity changes on output, this counterfactual distorts female labor supply choices to be based on their 1940 levels of $\bar{w}$ and $\bar{h}$, while actual wages and productivities are held at their 2000 levels. The resulting shift in labor supply is dramatic: the labor force participation rate of married women drops by 67% (unreported) – over twice as much as in the main counterfactual. As a result, market output falls by 14.1%, and total output declines by 7.7%. Comparing the 7.7% number to the 5.3% drop in total output in the main counterfactual suggests that distortions driven by cultural forces are almost 70% as consequential as those driven by economic forces.

Extensions and Robustness Figure 11 presents robustness checks and evaluates model extensions for the main counterfactual: the change in output in the 2000 economy had 1940 norms persisted.

Figure 11: Extensions and Robustness of Main Counterfactual



Note: This figure compares the market and total (market and nonmarket) output effects of various counterfactual scenarios, where the baseline year is set at 2000, with τ at 2000 being 0.15.

In case 2, wages adjust to changes in male and female labor supply under a CES production function where the two types of labor are imperfectly substitutable inputs.⁵¹ The output effects

⁵¹The elasticity of substitution between male and female labor is set to 3, following the estimation result in Acemoglu et al. (2004). Appendix H.2.1 describes how I solve for the equilibrium through fixed-point iteration.

remain close to the baseline because the estimated wage responses – a rise in female wages and a fall in male wages – are small.

Case 3 implements the counterfactual in the model extension with income effects.⁵² The results show smaller output effects than the baseline. This attenuation occurs because the change in expected economic utility within marriage decreases by less under this specification, which induces a smaller decline in marriage rates (-15%, compared to -26% in the baseline) and consequently a smaller reduction in education investment.

Cases 4 and 5 relax the baseline assumption that the observed wage differentials across groups – defined by sex, own education, spousal education if married, and children composition – reflect purely causal effects of belonging in that group. In the baseline, for example, an individual who marries is assumed to experience a change in market productivity equal to the observed difference in $\bar{w}$ between married and single individuals. However, the difference in $\bar{w}$ may also capture selection effects missing from the baseline model, such as sorting into marriage or education based on unobserved ability. If wage differentials are only partly causal, changes in productivity associated with changes in marital status or schooling in the baseline counterfactual may be overstated. To address this issue, I compress wage differentials based on standard assumptions or empirical decompositions from the literature that separate causal effects from selection. This provides a shortcut approach to netting out unmodeled selection effects, without explicitly modeling them. Details on the wage compression in cases 4 and 5 are provided in Appendix H.2.3.

In case 4, I assume a flat wage differential across marital status and partner type, adopting a common earnings equation specification (e.g., Chiappori et al., 2018) where earnings do not depend on these factors. This is a conservative specification that completely rules out the causal effect of marriage on productivity.⁵³ In light of the child penalty literature pioneered by Kleven et al. (2019), I still allow wages to differ by children composition index – reflecting skill depreciation when out of work, as in Adda et al. (2017), or reduced time availability and flexibility, as in Goldin (2014), due to children. The output effects are smaller than in the baseline, indicating that part of the decline in output in the baseline stems from the fact that the fall in marriage induces more individuals to receive single wages rather than married wages. Specifically, while single women’s $\bar{w}$ are found to be higher than married women’s $\bar{w}$ on average in 2000, the opposite is true for men. Thus, some of the baseline effect comes from more men receiving the lower single wages, which partially offsets the productivity gains from more women receiving the higher single wages. This component of the output effect is eliminated in case 4 by construction.

In case 5, I further compress the wage differential by education to 85% of its observed level,

⁵²See Appendix H.2.2 for details.

⁵³Korenman and Neumark (1991) find that only a small share of the marital wage premium of men is driven by selection on wages, wage growth, or other wage-enhancing characteristics, and instead provide support for the view that marriage itself enhances men’s labor market productivity.

in addition to the wage compression in case 4. The 85% compression represents a conservative benchmark, since calculations based on estimated education and ability premiums in [Chiappori et al. \(2018\)](#) show that the causal education premium typically accounts for a much larger share of the total wage gap by education.⁵⁴ As expected, the output effects are even smaller than in case 4, as the lower education investment in the counterfactual now translates to lower reductions in productivity.

Overall, the magnitude of market output loss ranges from -7.0% to -8.7% , while total output loss ranges from -4.0% to -5.3% . Even the smallest loss is roughly half the size of the output loss from labor supply distortions driven by economic forces (scenario 8, Figure 10). The qualitative finding of a substantial decline in both market and total output remains robust to alternative model assumptions.

6 Concluding Remarks

I measure and study the effects of gender roles associated with marriage on aggregate output, using historical data from the United States. Over the past 70 years, gender norms have become a weaker constraint on married women's labor supply, with the implied tax on their market wages falling from 44% to 14%.

Cultural constraints impose sizable output costs. Operating under 1940 norms wedges in the current day pulls down aggregate market output by 8.7% and aggregate total output by 5.3%. This amounts to nearly 70% of the output cost of the distortion in female labor supply from *economic* forces – when female labor force participation are based on wages and home productivities of 1940. The growth in norms wedges also account for 12.6% of the growth in market output and 8.1% of the growth in total output between 1940 and 2000. The effects are large, especially considering that these norms wedges capture a narrower concept of gender norms, relating to the distinction between married and single women who are otherwise similar, than the broader between-gender differences typically emphasized in the literature.

The magnitude of the output effects reflects the importance of endogenous adjustments in education and marriage markets, as counterfactual declines in output holding them fixed are significantly more modest. The amplification in general equilibrium occurs through multiple channels. First, stronger gender norms lower the marriage rate, especially among highly educated women. Given strong assortative matching by education, this weakens the marriage-market incentives to invest in education for both men and women. Second, for women, the reduced likelihood of labor force

⁵⁴Appendix I shows my calculations. Moreover, the empirical literature on returns to schooling, as surveyed in [Card \(1995\)](#), typically finds that instrumental variables (IV) estimates exceed ordinary least squares (OLS) estimates. This suggests that the education premium may be larger than observed wage differentials by education – implying an even greater fall in output in the main counterfactual than in the baseline.

participation after marriage diminishes the labor-market returns to education, even though fewer women marry. Third, changes in marriage and fertility patterns alter individuals' latent productivities.

My findings suggest that analyses that focus solely on immediate labor supply responses would substantially underestimate the true economic costs of gender norms, by missing the amplifying feedback effects through human capital investment and family formation decisions. Similarly, short-term evaluations of policy interventions or social movements that alter norms, or even perceptions of norms, would not capture the full impact that would emerge in the long term.

Despite its decline since 1940, the norms wedge remains well above zero, leaving scope for further output growth and gender convergence if cultural attitudes shift further. Yet its stagnation in recent decades suggests that additional forces – such as greater temporal flexibility in work arrangements (Goldin, 2014) – may be needed to enable married women to better combine market work with family responsibilities, which in turn could gradually foster further changes in norms.

The tractability of the model and transparent parameter inference procedure lend great practical advantages. For instance, Lee (2021) takes the model to 17 other countries and augments it with a male-female wedge, which is compared to the married women-single women wedge central to the current paper. On the other hand, there are other important dimensions that the model does not account for, such as occupations, divorce, childcare costs and transfers (Hannusch, 2024), and leisure (Aguiar and Hurst, 2007b). Building them in will allow a richer understanding of the effects of gender norms in marriage.

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Online Appendix

Appendix A Additional Tables

Table A1: Variation in Attitudes by Individual Characteristics

	Average	Shapley decomp (%)		Average	Shapley decomp (%)
	(1)	(2)		(3)	(4)
Education			Race		
None or grades 1-8	0.40		White	0.17	
Grades 9-11	0.28		Black	0.24	4.0
Grade 12	0.20	58.5	Other	0.21	
Some college	0.14				
College graduate or higher	0.09				
Survey year			Marital status		
1970-1979	0.25		Married	0.19	
1980-1989	0.16	12.5	Widowed	0.28	
1990-1999	0.16		Divorced	0.17	2.3
			Separated	0.19	
			Never married	0.15	
Number of children			Geographic region		
0	0.14		Northeast	0.17	
1	0.17		Midwest	0.18	1.9
2	0.18	10.1	South	0.21	
3	0.20		West	0.16	
4 or more	0.26				
Age			Sex		
20-29	0.15		Male	0.19	
30-39	0.16		Female	0.18	0.7
40-49	0.19	6.5			
50-59	0.24				

Note: This table reports how much various individual characteristics account for the variation in *attitude*, the indicator variable for an individual's disapproval of a married woman working in the labor market in the General Social Survey. The specific survey question is in Figure 3. Columns 1 and 3 report the weighted average disapproval rate by category of each variable, to show the variation across the categories. For columns 2 and 4, I regress *attitude* on the dummies for the categories of each variable and report the Shorrocks-Shapely decomposition denoting the relative contribution of each variable to the R squared.

Table A2: Years of Completed Schooling by Schooling Category by Year

Schooling level	Year						
	1940	1960	1970	1980	1990	2000	2010
1	[0,7]	[0,7]	[0,8]	[0,9]	[0,11]	[0,11]	[0,11]
2	8	8	[9,11]	[10,11]	12	12	12
3	[9,11]	[9,11]	12	12	[13,15]	[13,15]	[13,15]
4	12	12	[13,15]	[13,15]	16	16	16
5	[13,∞)	[13,15]	[16,∞)	16	[17,∞)	[17,∞)	[17,∞)
6		[16,∞)		[17,∞)			

Note: Years of completed schooling are integers in each interval. For example, individuals with 0,1,...,7 years of completed schooling fall under schooling level 1 in 1940.

Table A3: Description of children composition Categories

Family composition categories		Description
1		No child
2		1 child, aged 6-18
3		1 child, aged 0-5
4		2 or more children, all aged 6-18
5		2 or more children, at least one aged 0-5

Note: All children are one's own children living in the same household.

Table A4: Maximum Likelihood Estimate of θ

	1940	1960	1970	1980	1990	2000	2010
$\hat{\theta}$	1.55***	1.83***	1.84***	1.72***	1.98***	1.98***	1.84***
	(18.2)	(28.3)	(19.4)	(12.6)	(22.0)	(21.9)	(39.4)
N	156,404	315,190	315,424	562,479	699,598	796,976	736,666

Note: t statistics using standard errors clustered by sex in parentheses, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

When the standard errors are not clustered, following the model assumption of independently drawn market abilities, the t statistic is incredibly large due to the large sample size.

See step 2 of Section 4.2 for the maximum likelihood estimation strategy.

Table A5: Coefficients from Regressing $\bar{w}$ or $\bar{h}$ on Schooling Level

	Men		Women	
	$\bar{w}$	$\bar{h}$	$\bar{w}$	$\bar{h}$
Schooling level	1.555*** (0.373)	-0.017 (0.016)	1.340** (0.366)	0.039 (0.034)
Number of groups	1170	1170	1170	1170

Note: This table reports the coefficients on schooling level from regressions of group market wage ($\bar{w}$) and home productivity ($\bar{h}$) on schooling level, controlling for year, children composition, marital status, and spousal schooling level (all as dummy variables). Each observation corresponds to a group and is weighted by its empirical probability. As described in Appendix Table A2, schooling level is a categorical variable ordered by increasing years of completed schooling. For simplicity, it is treated as continuous in these regressions to illustrate how $\bar{w}$ and $\bar{h}$ vary with education. Standard errors (in parentheses) are clustered by census year to account for within-decade correlation due to decade-by-decade parameter inference and changing categorization of schooling level across census years. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

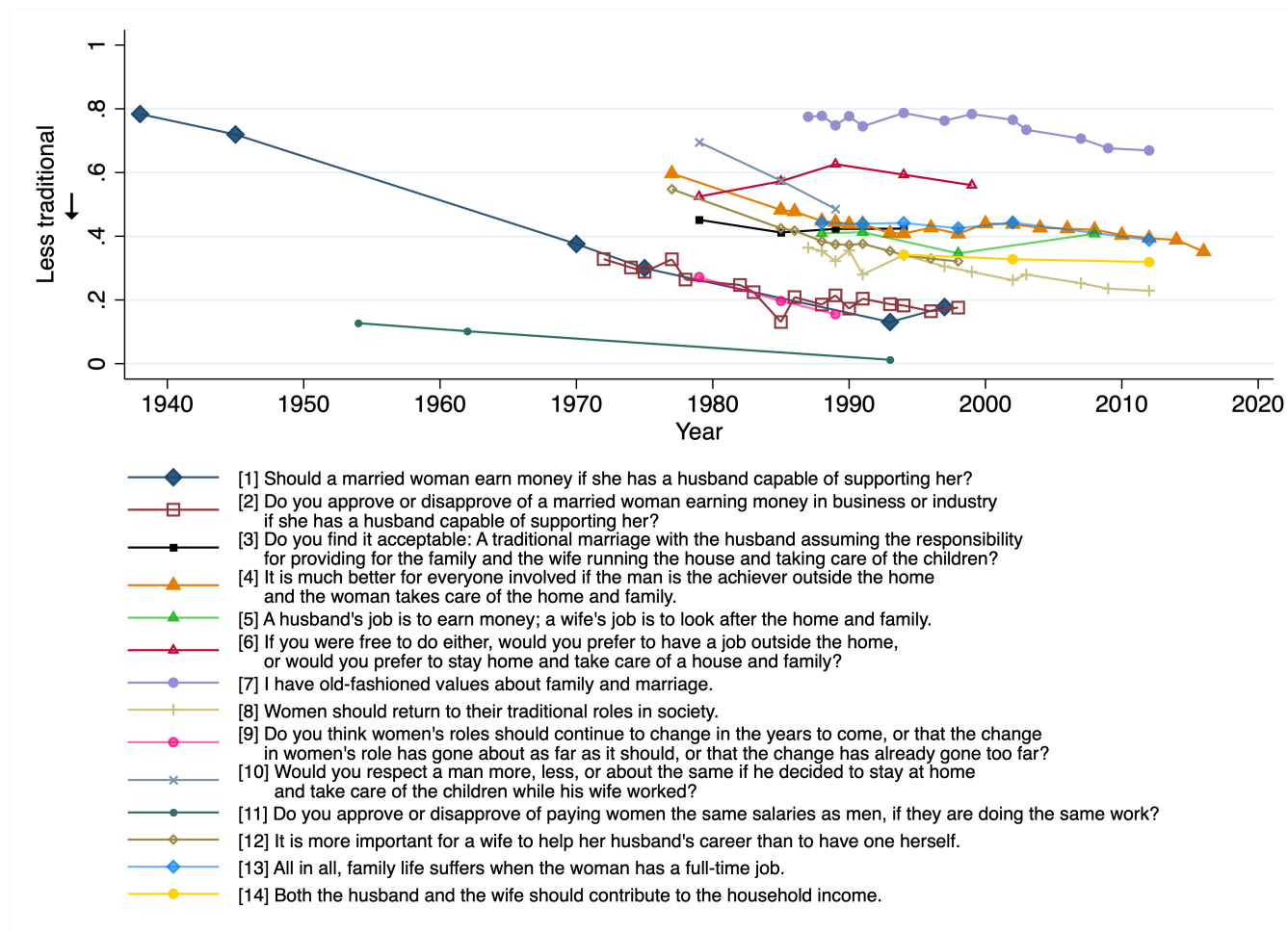
Table A6: Estimates of the Costs of Schooling c by Sex and Decade

		Schooling level					
		1	2	3	4	5	6
1940	Male	0.0	0.67	0.80	1.68	2.51	
	Female	0.0	1.06	1.05	1.98	3.53	
1960	Male	0.0	0.51	0.86	2.04	1.53	2.20
	Female	0.0	0.35	0.55	2.15	2.42	4.59
1970	Male	0.0	0.5	1.25	1.19	2.16	
	Female	0.0	0.02	1.17	2.00	4.60	
1980	Male	0.0	0.49	-0.09	-0.25	0.72	0.98
	Female	0.0	0.18	-0.17	0.47	2.13	2.91
1990	Male	0.0	-0.32	-0.22	0.93	2.51	
	Female	0.0	0.11	0.48	2.33	3.91	
2000	Male	0.0	-0.64	0.10	1.40	3.11	
	Female	0.0	-0.62	0.12	1.71	3.29	
2010	Male	0.0	-0.59	0.18	1.85	3.78	
	Female	0.0	-0.44	0.18	2.06	3.66	

Note: This table reports the estimates of c_g^s , the utility cost of schooling by sex and decade. As a normalization, I set $c_g^1 = 0$. The schooling categories for each decade are described in Appendix Table A2.

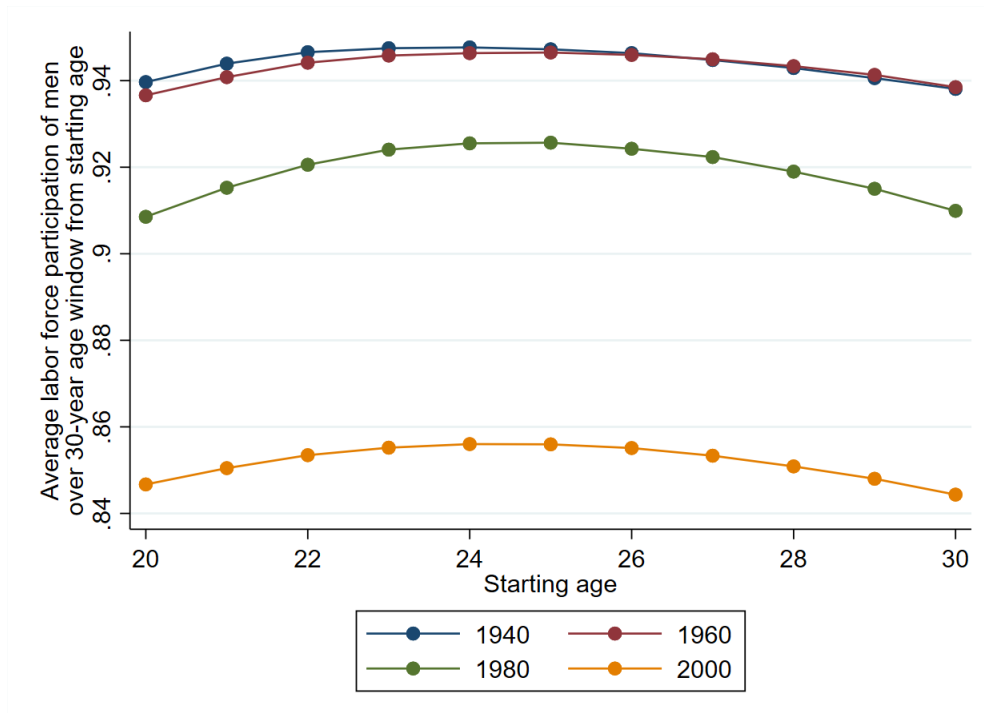
Appendix B Additional Figures

Figure A1: Trends of Attitudes Relating to Gender Roles within Marriage



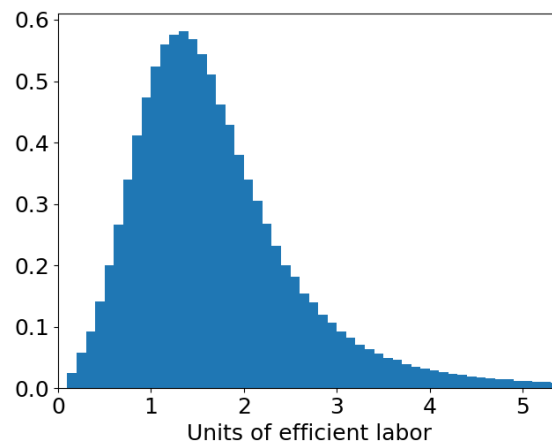
Note: The attitudinal survey data originates from the ROPER polls database and the General Social Survey. For ROPER polls, I perform a keyword search of “women” and “gender,” select the questions relating to gender roles within marriage, and among them, select the ones asked repeatedly over multiple periods. All samples are nationally representative adults or nationally representative men and women separately. All survey answers have been reconfigured to lie between 0 and 1, with 1 corresponding to the most traditional answer and 0 the least. This figure plots the weighted averages of the reconfigured answers by year.

**Figure A2: Average Labor Force Participation of Men
over a 30-Year Window from Starting Age**



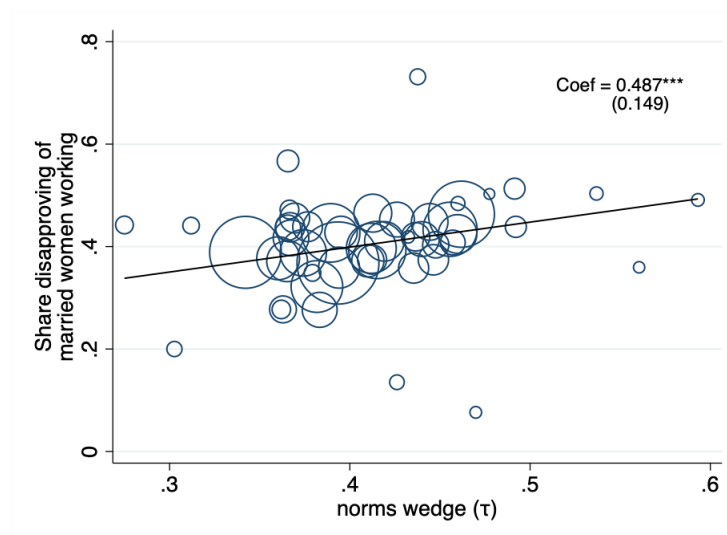
Note: This figure plots the weighted average of labor force participation among men aged between starting age and (starting age+29). It shows that 25-54 is an appropriate age range for the economically active years of one's life and that this observation is quite stable over time.

Figure A3: Histogram of Empirical Market Abilities



Note: This figure plots the histogram of the market abilities of all working individuals with wage data, where the market abilities are inferred using the model structure as outlined in step 2 of Section 4.2.

Figure A4: Conservative states have higher norms wedges

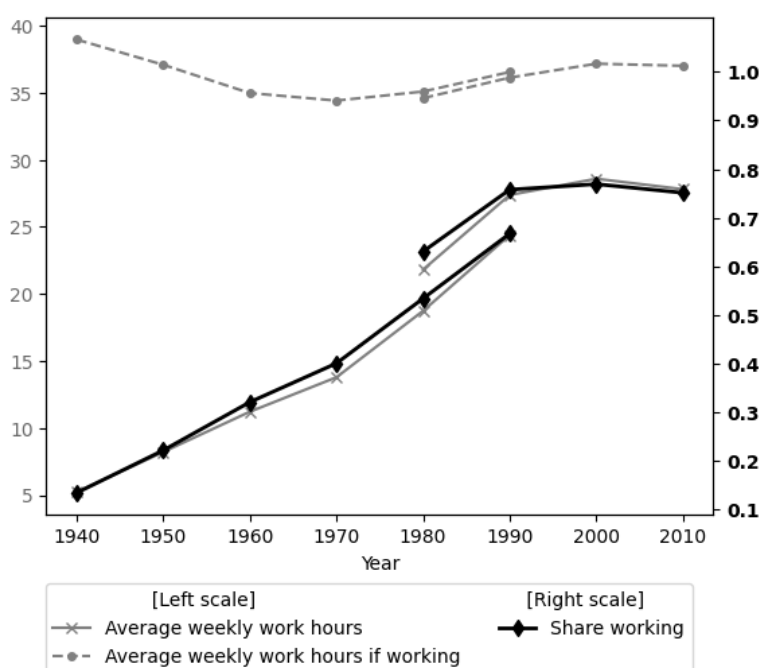


Note: This figure visually shows the result in column 1 of Table 1: states with higher τ are also the ones where a greater share of individuals disapprove of married women working. Each observation is a state, and the bubble size represents the state population size.

Appendix C Growth of Married Women's Market Work

Figure A5 illustrates that the growth in married women's market work between 1940 and 2010 was driven by growth along the extensive margin of labor force participation. It presents the average weekly market work hours of married women, both conditional and unconditional on working, as well as the share of married women working. Conditional work hours (intensive margin) remain stable over time, while the share working (extensive margin) closely tracks the rise in unconditional hours (total growth).

Figure A5: Extensive vs. Intensive Margin of Married Women's Market Work



Note: This figure presents the average weekly market work hours of married women, shown both conditional on working positive hours (circles, dashed lines) and unconditional (crosses, thin solid lines), along with the share of married women working positive hours (diamonds, thick solid lines). Because no single measure of weekly market hours is available across all decades, I plot two overlapping variables to demonstrate that the trend is consistent regardless of the measure used. From 1940 to 1990, I use the midpoint of the interval variable *hrswork2* (hours worked last week, interval), and from 1980 to 2010, I use *uhrswork* (usual hours worked per week).

In addition to the visual illustration, I formally calculate the contribution of the extensive margin in driving the growth in unconditional work hours.

Average market work hours equals the product of the share of working individuals (extensive margin) and the average hours worked conditional on working (intensive margin):

Where W is the set of individuals who are working,

$$\overline{workhrs} = \frac{\sum_i workhrs_i}{N} = \frac{\sum_{i \in W} workhrs_i}{N} = \frac{|W|}{N} \times \frac{\sum_{i \in W} workhrs_i}{|W|} = \underbrace{\overline{work}}_{\text{extensive}} \times \underbrace{(\overline{hrs|work})}_{\text{intensive}}$$

Thus, the growth in married women's work hours from period s to t can be decomposed into the extensive and intensive margins:

$$\overline{workhrs}_t - \overline{workhrs}_s = \underbrace{(\overline{work}_t - \overline{work}_s) \times \overline{hrs|work}_s}_{\text{extensive margin}} + \underbrace{(\overline{hrs|work}_t - \overline{hrs|work}_s) \times \overline{work}_t}_{\text{intensive margin}}$$

The contribution of the extensive margin to the growth over 1940-2010, then, can be calculated in the following ways.

1. Long-difference: $t = 2010, s = 1940$.

Using a harmonized measure of annual work hours (weekly work hours $\times$ number of weeks worked in Table A8), the result is

$$\frac{(\overline{work}_{2010} - \overline{work}_{1940}) \times \overline{hrs|work}_{1940}}{\overline{workhrs}_{2010} - \overline{workhrs}_{1940}} = 99.1\%$$

2. Total decadal-difference:

Different measures of weekly work hours and the number of weeks worked are used across years to construct the harmonized measure of annual work hours. To circumvent the measurement inconsistency in 1940 and 2010, I also examine the total decadal-difference that uses the same measure within each decadal-difference pair. I use the midpoint of the intervalled variable *hrswork2* (hours worked last week, intervalled) for 1940-1990, and *uhrswork* (usual hours worked per week) for 1980-2010. For 1980-1990, when both variables are available, I take the average of the decadal-difference from each variable. The result is

$$\frac{\sum_{t=1950}^{2010} (\overline{work}_t - \overline{work}_{t-10}) \times \overline{hrs|work}_{t-10}}{\sum_{t=1950}^{2010} \overline{workhrs}_t - \overline{workhrs}_{t-10}} = 96.0\%$$

Therefore, the growth of married women's market work from 1940 to 2010 is almost entirely driven by the extensive margin.

Appendix D Model

D.1 General form of the utility function

The general form of the utility function is given by

$$u_i(Q, C_i, L_f, L_m) = H\left(f(Q)C_i - r(Q)\tau_i w_f L_f + g_i(Q)\right)$$

where the following conditions hold:

Conditions

C1) H is strictly increasing and strictly concave

C2) $(H')^{-1}$ is homogeneous or logarithmically homogeneous^{55,56}

C3) $2p(f')^2 - p \cdot f \cdot f'' + \tau w_f L_f (r'' f' - r' f'') - f' g'' + g' f'' > 0$, where $\tau \equiv \tau_m + \tau_f$, and $g(Q) \equiv g_m(Q) + g_f(Q)$

This general utility function yields the same result that 1) the optimal labor supply decision of the couple is one that maximizes their pooled income less the disutilities from the wife working in the labor market, and 2) the optimal labor decisions are made independently based only on individuals' comparisons of the gains from working in the market versus at home.

⁵⁵Condition C2) is based on [Mazzocco \(2004\)](#), which shows that a collective household's behavior under uncertainty is equivalent to that of a representative agent if and only if H is of the Identically Shaped Harmonic Absolute Risk Aversion (ISHARA) class:

$$-\frac{H_i''(x)}{H_i'(x)} = \frac{1}{\theta x + a_i}$$

⁵⁶CRRA ($H(x) = \ln(x)$, or $H(x) = \frac{x^{1-\theta}-1}{1-\theta}$ for $\theta > 0, \theta \neq 1$) and CARA ($H(x) = -\exp\{-\theta x + b\}$ for $\theta > 0, b \in R$) utility functions – the most commonly used utility functions for risk-averse individuals – satisfy condition C2).

D.2 Model discussion: How home production features as “income” in the budget constraint

It might seem non-standard that according to the budget constraint, private and public goods can be bought with “income” from home production. However, the maximization problem is equivalent to solving

$$\begin{aligned} \max_{Q,C,Y,B} & (Q + Y)(C + D - \tau w_f L_f) \\ \text{s.t.} \quad & pQ + C = w_m L_m + w_f L_f \\ & pY + B = h_m(1 - L_m) + h_f(1 - L_f) \end{aligned}$$

where Y is the non-rival, public component of home production (e.g. cleaning communal area, childcare) and $D \equiv D_m + D_f$ is the total consumption of the private component of home production (e.g. cleaning private space, laundering own clothes). Here, market goods and services Q and C can only be financed from market earnings, while consumption of home-produced goods and services occur within the total home production done by the couple. The market value of private home goods is normalized to be the same as that of the private market good (the numeraire), and the market value of public home goods is the same as the price of public market good.

D.3 Model discussion: value of staying home beyond housework

Following [Gronau \(1977\)](#), I can categorize home production into work at home, which is perfectly substitutable to work in the market, and leisure (i.e. home consumption time), which has poor market substitutes. Then, the utility function – in the general form as in [Appendix D.1](#) – can be formulated as

$$u_i(Q, C_i, L_f, L_m) = H\left(f(Q)C_i - r(Q)[a_i w_i L_i + \tau_i w_f L_f] + g_i(Q)\right)$$

where a_i denotes i ’s preference for leisure/home consumption time, measured proportionally to i ’s market wage.⁵⁷

Ultimately, the optimal labor supply decisions would come down to

$$\begin{aligned} L_m^* &= \mathbb{1}[w_m - a_m \geq h_m] \\ L_f^* &= \mathbb{1}[(1 - \tau)w_f - a_f \geq h_f] \end{aligned}$$

⁵⁷It is easy to also introduce j valuing i ’s home consumption time, e.g. the husband valuing the wife’s play time with their children.

D.4 Model extension: Enriching women's labor supply decisions

The independence in the labor supply decisions of husbands and wives simplifies the study of these decisions.⁵⁸ However, a necessary assumption to generate the independence is the perfect substitutability of home-produced goods and market goods. The perfect substitutability assumption, in turn, implies that there are no incentives for specialization. I can enrich the labor supply decisions of married women, however, if I abstract from the labor supply decisions of married men, i.e. have them always work in the market.

As in [Aguiar and Hurst \(2007a\)](#), consider the final consumption good being produced according to a CES production function with market goods and home production as inputs:

$$\left[(y + wL)^\rho + (\underline{h} + h(1 - L))^\rho \right]^{\frac{1}{\rho}} \quad (23)$$

y refers to nonlabor household market income, including any remittances from family, government benefits, and, if married, the husband's earnings. $\underline{h}$ refers to the basic level of home production conducted in the household, including the amount of home production a woman would have completed regardless of her labor force participation, and again, if married, any home production completed by the husband. L is the labor force participation status of single men and women for single households, and of wives for married households.

To make use of the convenient properties of the Fréchet-distributed market and home abilities, a simplification is necessary. A first-order Taylor approximation of equation (23) around $(w = 0, h = 0)$ yields

$$(y^\rho + \underline{h}^\rho)^{\frac{1}{\rho}} + (y^\rho + \underline{h}^\rho)^{\frac{1-\rho}{\rho}} y^{\rho-1} wL + (y^\rho + \underline{h}^\rho)^{\frac{1-\rho}{\rho}} \underline{h}^{\rho-1} h(1 - L) \quad (24)$$

Thus, equation (24) replaces the right-hand side of the budget constraints in household utility maximization problems (equations (2) for married households and (7) for single households). The optimal labor supply decision for single men and women maximizes equation (24):

$$\hat{L}^* = \mathbb{1} [y^{\rho-1} w \geq \underline{h}^{\rho-1} h]$$

A married woman, on the other hand, must consider the cost of the norms wedge. Where the cost of working in the market is denominated proportionally to the contribution of market goods in

⁵⁸The independence assumption allows me to take advantage of the convenient properties of the Fréchet distribution to model selection into market work versus home production, separately for men and women. This structure also enables model extensions such as in [Appendix D.5](#), where I add in gender norms wedges for married men.

the production of the final good,⁵⁹ her optimal labor supply decision is:

$$L^* = \mathbb{1} \left[(1 - \tau)y^{\rho-1}w \geq \underline{h}^{\rho-1}h \right] \quad (25)$$

The norms wedges from this extension, comparable to ones in the benchmark model given by equation (20), is:

$$\tau^{qr}(\mathcal{K}) = 1 - \frac{\text{avgwage}_F^{0r}(\mathcal{K})}{\text{avgwage}_F^{qr}(\mathcal{K})} \cdot \underbrace{\left(\frac{\underline{h}_F^{0r}}{\underline{h}_F^{qr}} \cdot \frac{y_F^{qr}}{y_F^{0r}} \right)^{1-\rho}}_{\text{new in the extension}} \cdot \left(\frac{1 - P_F^{0r}(\mathcal{K})}{1 - P_F^{qr}(\mathcal{K})} \right)^{\frac{1}{\theta}} \quad (26)$$

The following describes how I compute the new part of the norms wedge in this extension.

- ρ : set at 0.5, since the elasticity of substitution between market goods and home produced goods in the literature is estimated at around 2; [McGrattan et al. \(1997\)](#): 1.8, [Chang and Schorfheide \(2003\)](#): 2.3, [Rupert et al. \(1995\)](#): 1.8, [Aguiar and Hurst \(2007a\)](#): 1.78-2.13.
- y : the sum of welfare benefits, social security income, other unearned income, and, if married, spousal earnings in the U.S. Decennial Census. The values are truncated below at 0 and winsorized at the top 10% to remove distortions from outliers, and then I take the average by group. I compute (26) only from 1960 because information on non-wage income becomes available from 1960.
- $\underline{h}$: In the benchmark model, single and married women with the same level of education share the same home productivity. Similarly, I assume here that $\frac{\underline{h}_F^{0r}}{\underline{h}_F^{qr}}$ is a constant. As the aim of computing the norms wedge in the extension is to compare its evolution trajectory over time to that of the wedges in the benchmark model, the value of this constant is set to equate the norms wedges in the benchmark and the extension in 1960.

⁵⁹The married individual i 's utility in equation (1) becomes $\ln(Q) + \ln(C_i - \tau_i(y^\rho + \underline{h}^\rho)^{\frac{1-\rho}{\rho}} y^{\rho-1} w_f L_f)$.

D.5 Model extension: Adding male norms wedges

The independence in the labor supply decisions of husbands and wives, arising from the perfect substitutability of home-produced goods and market goods, allows for the addition of *male* norms wedges. A married couple receives disutility from not only the wife working in the market, but also the husband working in home production. Therefore, married individual $i \in \{m, f\}$ gets the following utility:

$$u_i(Q, C_i, L_f, L_m) = \ln(Q) + \ln(C_i - \tau_i w_f L_f - \tilde{\tau}_i h_m (1 - L_m))$$

where $\tilde{\tau}_i$ represents the disutility that i gets from the husband m working. With $\tilde{\tau} \equiv \tilde{\tau}_m + \tilde{\tau}_f$, the optimal labor supply decisions are:

$$\begin{aligned} L_m^* &= \mathbb{1}[w_m \geq (1 - \tilde{\tau})h_m] \\ L_f^* &= \mathbb{1}[(1 - \tau)w_f \geq h_f] \end{aligned}$$

D.6 Derivation of marriage market equilibrium

The probability that man m chooses spousal type $r \in \{1, \dots, S\}$ or stays single ($r = 0$) is

$$\mathbb{P}(r = \arg \max_{r'=0,1,\dots,S} V_m^{qr'}) = \frac{\exp\{\mathbb{E}(v_m^{qr}) - \pi^{qr} + \psi^{qr}\}}{\sum_{r'=0}^S \exp\{\mathbb{E}(v_m^{qr'}) - \pi^{qr'} + \psi^{qr'}\}}$$

The maximum likelihood estimator of $\mathbb{P}(r = \arg \max_{r'=0,1,\dots,S} V_m^{qr'})$ is the fraction of type q men married to r , or $\frac{n^{qr}}{M^q}$.

Hence, in terms of the number of (q, r) marriages *demanded* by type q men,

$$\ln n^{qr, \mathcal{D}} = \ln n^{q0} + \mathbb{E}(v_m^{qr}) - \pi^{qr} + \psi^{qr} - \mathbb{E}(\hat{v}_m^q)$$

Similarly, woman f of type r choosing her spousal type or remaining single gives the analogue for the number of (q, r) marriages *supplied* by type r women,

$$\ln n^{qr, \mathcal{S}} = \ln n^{0r} + \mathbb{E}(v_f^{qr}) + \pi^{qr} + \psi^{qr} - \mathbb{E}(\hat{v}_f^r)$$

The marriage market-clearing equilibrium transfers π^{qr} are determined such that

$$n^{qr, \mathcal{D}} = n^{qr, \mathcal{S}} = n^{qr}.$$

Appendix E Data

U.S. Census samples The primary data source is the U.S. Decennial Census (Ruggles et al., 2024), accessed through IPUMS USA on October 29, 2024. IPUMS provides multiple samples for each census year. Table A7 indicates the specific sample used for each decade. I use the “default” sample for each year as designated by IPUMS USA. When multiple default samples are available, I select the largest one. However, if the chosen sample exceeds 1% of the population, I randomly draw observations so that each year corresponds to 1% of that year’s population, to ensure consistent representation across years in the pooled data.

Table A7: Raw IPUMS USA Samples Used for each Decade

1900	1%	1930	1%	1960	5%	1990	5% state
1910	1%	1940	1%	1970	1% metro fm2	2000	5%
1920	1%	1950	1%	1980	5% state	2010	ACS 2010, 2011, 2012

Notes: The sample names follow those listed on the IPUMS “Select Samples” page.

Measuring market wages The “market wage” in this paper is defined as each respondent’s total pre-tax wage and salary income for the previous year (variable *incwage*) divided by annual work hours. *incwage* is deflated using the price index variable *CPI99*, so the units are in 2000 US dollars.

Annual market work hours are calculated as the product of weekly work hours and the number of weeks worked, with the underlying variables listed in Table A8. “Usual hours worked per week” is used over “hours worked last week,” if available. Continuous measures are used whenever available. When only interval measures exist, I convert them to continuous values using the corresponding interval averages from other years.

Table A8: Raw Variables for Measuring Annual Market Work Hours by Year

	1940, 1950	1960, 1970	1980, 1990, 2000	2010
Weekly work hours	hrswork1	hrswork2	uhrswork	uhrswork
Number of weeks worked	wkswork1	wkswork2	wkswork1	wkswork2

Notes: *hrswork1*: Hours worked last week; *hrswork2*: Hours worked last week, intervalled; *uhrswork*: Usual hours worked per week; *wkswork1*: Weeks worked last year; *wkswork2*: Weeks worked last year, intervalled.

Appendix F Parameter Inference

F.1 Derivation of the likelihood function to estimate θ

The probability density function of the Fréchet distribution with shape parameter θ and scale parameter s is:

$$f(x; s, \theta) = \frac{\theta}{s} \exp \left\{ - \left(\frac{x}{s} \right)^{-\theta} \right\} \left(\frac{x}{s} \right)^{-\theta-1}$$

The scale parameter in the Fréchet distribution of market abilities among market workers is $\left(\frac{1}{P}\right)^{\frac{1}{\theta}}$, where P is the fraction working in the market among a group defined by gender g , match (q, r) , and children composition $\mathcal{K}$. Therefore, where x_n is the market ability of observation n and P_n denotes the fraction of workers in observation n 's group, the maximum likelihood estimator for θ is:

$$\tilde{\theta}_{MLE} = \arg \max_{\theta \in (0, \infty)} \sum_{n=1}^{Nobs} [\ln \theta - \ln P_n - x_n^{-\theta} P_n^{-1} - (\theta + 1) \ln x_n]$$

F.2 Extracting market abilities from market wages

Recall that an individual i of gender g in a (q, r) match with children composition $\mathcal{K}$ receives wage

$$w_{it} = \bar{w}_{gt}^{qr}(\mathcal{K}) \epsilon_i^w, \quad g \in \{M, F\}$$

where $\bar{w}_{gt}^{qr}(\mathcal{K})$ is the market wage per unit of effective labor for each group. Therefore the x_n in practice equals ϵ_{it}^{w*} where the asterisk denotes that this is the market ability of those who choose to work in the market. Let us isolate ϵ_{it}^{w*} from the observed wages:

$$\text{logwage}_{it} = \ln \bar{w}_{gt}^{qr}(\mathcal{K}) + \ln \epsilon_{it}^{w*}$$

To this end, I regress log wages on (decade $\times$ sex $\times$ education pair $\times$ children composition) dummies. For each group, then, the residuals are

$$\begin{aligned} \text{residuals}_{it} &= \text{logwage}_{it} - \overline{\text{logwage}_{it}} \\ &= (\ln \bar{w}_{gt}^{qr}(\mathcal{K}) + \ln \epsilon_{it}^{w*}) - (\ln \bar{w}_{gt}^{qr}(\mathcal{K}) + \overline{\ln \epsilon_{it}^{w*}}) \\ &= \ln \epsilon_{it}^{w*} - \overline{\ln \epsilon_{it}^{w*}} \end{aligned}$$

Thus,

$$\begin{aligned}
x_n &= \exp\{\text{residuals}_{it} + E(\ln \epsilon_{it}^{w*})\} \\
&= \exp\{\text{residuals}_{it} + \frac{\gamma}{\theta} + \ln(s_n)\} \\
&= \exp\{\text{residuals}_{it} + \frac{\gamma}{\theta} - \frac{1}{\theta} \ln P_n\}
\end{aligned}$$

where γ is the Euler's constant.⁶⁰

F.3 Reformulation of the expected utility from each schooling level

From the marriage market clearing condition and equation (4), equilibrium transfers are given by

$$\begin{aligned}
\pi^{qr} &= \frac{\ln n^{q0} - \ln n^{0r} - \ln \mu^{qr} - \mathbb{E}(\hat{v}_m^q) + \mathbb{E}(\hat{v}_f^r)}{2} \\
&= \frac{\ln n^{q0} - \ln n^{0r} - \ln \mu^{qr} - 2\mathbb{E}(\ln \hat{Q}_m^q) + 2\mathbb{E}(\ln \hat{Q}_f^r)}{2}
\end{aligned}$$

Then,

$$\begin{aligned}
U_F^r &= \sum_{q=0}^S \left[\frac{n^{qr}}{F^r} \left(\mathbb{E}(v_f^{qr}) + \pi^{qr} + \psi^{qr} \right) \right] - c_F^r - \xi^r \\
&= \sum_{q=0}^S \left[\frac{n^{qr}}{F^r} \left(2\mathbb{E}(\ln Q^{qr}) + \ln \frac{\mu^{qr}}{1 + \mu^{qr}} + \pi^{qr} + \psi^{qr} \right) \right] + \ln p - c_F^r - \xi^r \\
&= \sum_{q=0}^S \left[\frac{n^{qr}}{F^r} \left(2\mathbb{E}(\ln Q^{qr}) + \frac{1}{2} \ln \mu^{qr} - \ln(1 + \mu^{qr}) + \frac{\ln n^{q0} - \ln n^{0r}}{2} - \mathbb{E}(\ln \hat{Q}_m^q) + \mathbb{E}(\ln \hat{Q}_f^r) + \psi^{qr} \right) \right] \\
&\quad + \ln p - c_F^r - \xi^r \\
&= \sum_{q=0}^S \left[\frac{n^{qr}}{F^r} \left(2\mathbb{E}(\ln Q^{qr}) + \Psi^{qr} + \frac{\ln n^{q0} - \ln n^{0r}}{2} - \mathbb{E}(\ln \hat{Q}_m^q) + \mathbb{E}(\ln \hat{Q}_f^r) \right) \right] + \ln p - c_F^r - \xi^r \\
&= \sum_{q=0}^S \left[\frac{n^{qr}}{F^r} \left(2A^{qr} + \Psi^{qr} + \frac{\ln n^{q0} - \ln n^{0r}}{2} - \hat{A}_M^q + \hat{A}_F^r \right) \right] + \ln p - c_F^r - \xi^r
\end{aligned}$$

⁶⁰To compute $E(\ln \epsilon_{it}^{w*})$, I need the probability density function of $y = g(x) = \ln(x)$ where x is a Fréchet random variable with scale parameter s and shape parameter θ . $f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right| = \frac{\theta}{s} \left(\frac{e^y}{s} \right)^{-\theta-1} e^{-\left(\frac{e^y}{s} \right)^{-\theta}} |e^y|$. Thus, $E(y) = \theta s^\theta \int_{-\infty}^{\infty} y e^{-\theta y} e^{-e^{-\theta y} s^\theta} dy = -\frac{1}{\theta} \int_0^{\infty} e^{-z} (\ln z - \theta \ln s) dz = -\frac{1}{\theta} \Gamma'(1) + \ln s = \frac{\gamma}{\theta} + \ln s$ where γ is the Euler's constant.

It can be shown similarly that

$$\begin{aligned}
U_M^q &= \sum_{r=0}^S \left[\frac{n^{qr}}{M^q} \left(\mathbb{E}(v_m^{qr}) - \pi^{qr} + \psi^{qr} \right) \right] - c_M^q - \xi^q \\
&= \sum_{r=0}^S \left[\frac{n^{qr}}{M^q} \left(2\mathbb{E}(\ln Q^{qr}) + \Psi^{qr} - \frac{\ln n^{q0} - \ln n^{0r}}{2} + \mathbb{E}(\ln \hat{Q}_m^q) - \mathbb{E}(\ln \hat{Q}_f^r) \right) \right] + \ln p - c_M^q - \xi^q \\
&= \sum_{r=0}^S \left[\frac{n^{qr}}{M^q} \left(2A^{qr} + \Psi^{qr} - \frac{\ln n^{q0} + \ln n^{0r}}{2} + \hat{A}_M^q - \hat{A}_F^r \right) \right] + \ln p - c_M^q - \xi^q
\end{aligned}$$

Appendix G Norms Wedges under Alternative Models

G.1 Male norms wedges

What if married *men* are also subject to norms wedges? The male counterpart to the homemaker gender role would be the breadwinner gender role where married men are expected to earn a market income. This situation corresponds to the model extension in Appendix D.5. Because the optimal labor supply decisions of husbands and wives are independent, the values of the norms wedges applying to married women are not affected. Nonetheless, it would be interesting to compare the “female” norms wedges to the “male” norms wedges.

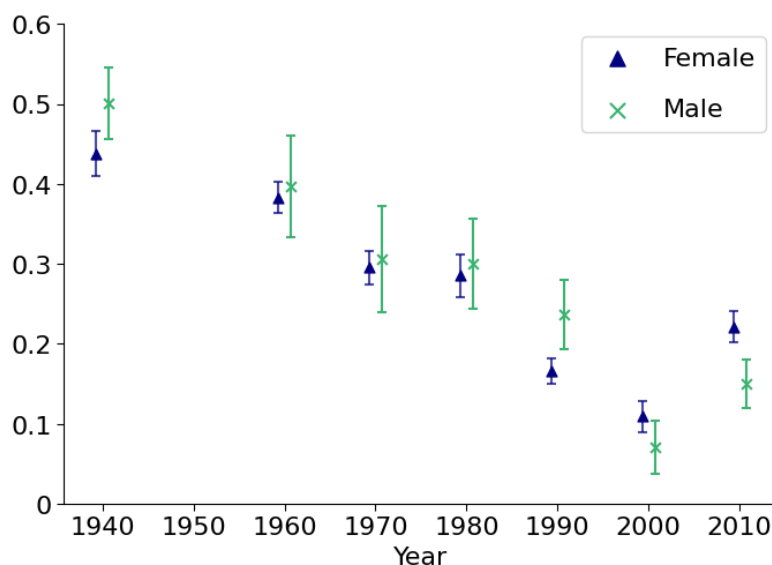


Figure A6: Female and Male Norms Wedges by Decade

Note: This figure plots the weighted median of τ and $\tilde{\tau}$ (described in Appendix D.5), inferred for each group, by decade. The weight equals the empirical probability of each group. The error bars indicate 95% confidence intervals based on bootstrapped standard errors with 50 replications.

Figure A6 shows that the male norms wedge declines over time, resembling the trend of the female norms wedge. The decline indicates that the labor force participation of married men are falling faster than that of comparable single men. In fact, the male norms wedge is even higher than the female counterpart in 1940 and is lower towards the end of the sample period. This result may be counterintuitive; it is difficult to reconcile the consistent and large decline with the fact that the rise of the stay-at-home fathers is only a recent phenomenon.

However, the male norms wedges must be interpreted with a grain of salt. The first reason is an algebraic one. The male norms wedges are proportional to the ratio of *nonparticipation in the labor force* of married and single men.⁶¹ Throughout the sample period, the male labor force participation rate is very close to 1 regardless of marital status, so that small changes in the labor force participation rate translate to large changes in norms wedges. For the same reason, the male norms wedge is also much noisier than the female counterpart, as depicted by the wider confidence intervals.⁶² The second reason is that nonparticipation in the labor force is less synonymous with home production for men, so $\tilde{\tau}$ would have less to do with actual gender norms. Indeed, state-level male norms wedges are not correlated to state-level attitudes towards gender roles, unlike female norms wedges (Table 1).

G.2 Robustness of norms wedges

Figure A7 demonstrates the robustness of the values of the norms wedges to alternative model specifications and sample selection criteria.

“Baseline” refers to the norms wedges in the baseline model, reproducing the norms wedges in Figure 7. The rest are described as follows.

- (1) I incorporate differential tax treatment by marital status by using imputed after-tax wages instead of gross wages. A married woman’s labor supply decision may differ from that of an otherwise similar single woman due to access to more favorable tax arrangements. Because after-tax earnings are not recorded in the U.S. Decennial Census, I apply federal nominal income tax rates from [Tax Foundation \(2013\)](#) to reported before-tax earnings, assuming individuals file under the most beneficial category: Married Filing Jointly, Married Filing Separately, Single, or Head of Household.
- (2) In the baseline, market wages are calculated as wage and salary income divided by annual market work hours (see Appendix E). Here, I augment income by including net pre-tax business, professional practice, and farm income (variable *incbusfm*), available from 1960 onward.
- (3) In the baseline, the Fréchet shape parameter θ , which governs the dispersion of idiosyncratic

⁶¹The male norms wedge for a married man in a (q, r) match with children composition $\mathcal{K}$ is

$$\tilde{\tau}^{qr}(\mathcal{K}) = 1 - \frac{\text{avgwage}_M^{qr}(\mathcal{K})}{\text{avgwage}_M^{q0}(\mathcal{K})} \left(\frac{1 - P_M^{qr}(\mathcal{K})}{1 - P_M^{q0}(\mathcal{K})} \right)^{\frac{1}{\theta}}$$

⁶²The noisiness can be viewed as a consequence of the difficulty of estimation when the parameter is near the boundary.

market and home abilities, is estimated separately for each decade. Here, I estimate a constant θ using pooled data from 1940–2010, following [Hsieh et al. \(2019\)](#), implying that the distribution of innate talent does not change over time.

- (4) The baseline defines married couples as those with a “probable” spouse present in the household, including cohabiting partners, as identified by IPUMS. Here, I adopt a stricter definition, restricting married couples to those in legal marriages – specifically, individuals whose marital status is “married, spouse present in household.”
- (5) To assess the sensitivity of the norms wedges to the composition of singles, I exclude widows and widowers from the group of single individuals. Widowed singles are a category of interest, as widowhood is more exogenously determined than never marrying or separating.
- (6) A married woman’s labor supply decision may differ from that of a single woman due to higher future childbearing prospects, even when their current children composition index is the same. Childbearing prospects affect a woman’s decision to work, as the expectation of fewer career interruptions increases the incentive to participate in the labor market. Employers may also be less willing to hire women they perceive as likely to leave temporarily or permanently in the future. To account for this difference, I limit the sample to individuals aged 40–54, past the prime childbearing age, rather than 25–54 in the baseline.
- (7) The schooling levels vary by decade in the baseline (Appendix Table [A2](#)), balancing the need for detailed categories to ensure comparability between married and single women with the need for sufficient sample sizes to reduce noise in the estimated norms wedges. Here, I test robustness to using a fixed set of education levels across all decades. To ensure sufficient sample sizes in each decade, they are: less than high school, high school graduate, and more than high school.

Across these alternative specifications, the magnitude and trends of the estimated norms wedges remain consistent, providing reassurance about the robustness of the baseline results.

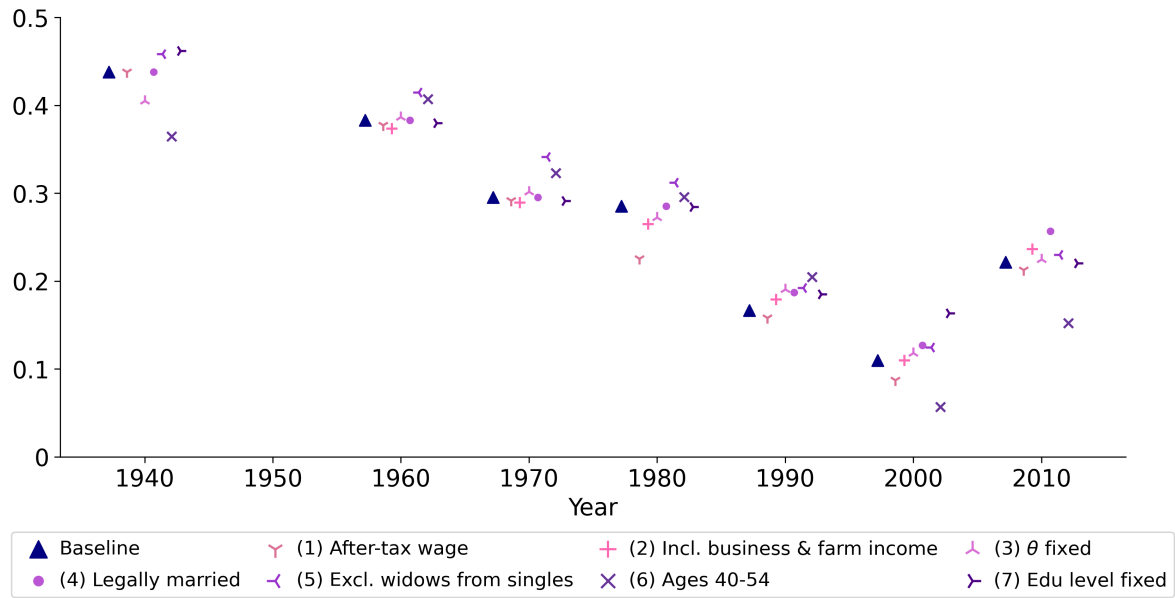


Figure A7: Norms Wedges Under Alternative Specifications or Sample Criteria

Note: This figure plots the values of the norms wedges by decade, under alternative model specifications or sample criteria.

Appendix H Counterfactual Steps and Robustness

H.1 Steps for conducting counterfactuals

I denote counterfactual values with underlines.

1. Compute $\underline{A}^{qr}$ and $\underline{P}_g^{qr}(\mathcal{K})$. They differ from the values at the status quo because optimal labor decisions would change under different gender norms. Note that $\hat{A}_g^s$ remains unchanged, as singles' labor decision is unaffected by norms wedges.
2. Solve for the number of marriage matches $\underline{n}^{qr}$ and education shares $\{\underline{M}^q, \underline{F}^r\}$, by substituting $\underline{A}^{qr}$ with $\underline{A}^{qr}$ in the marriage-market equation (21) and education equation (22). I also use the accounting identities given by equations (12) and (13), and assume that total population size is fixed.
3. Finally, compute $\underline{Y}^{mkt}$ and $\underline{Y}^{home}$.

H.2 Robustness of the main counterfactual

This section describes the different cases in Figure 11, which shows how the results of the main counterfactual (change in aggregate output in 2000 with 1940 norms wedges) change under alternative assumptions.

H.2.1 Case 2: CES Production Function with Male and Female Labor

Assume the market production function is no longer linear in male and female labor, but instead takes the Constant Elasticity of Substitution (CES) form, as in Acemoglu et al. (2004). Following their estimation results, the elasticity of substitution, $1 - \frac{1}{r}$, is set to 3: male and female labor are close, but far from perfect, substitutes.

$$Y^{mkt} = B [(1 - \lambda)\mathcal{M}^r + \lambda\mathcal{F}^r]^{1/r} \quad (27)$$

The male and female wages are given by

$$\begin{aligned} w_M &= B [(1 - \lambda)\mathcal{M}^r + \lambda\mathcal{F}^r]^{\frac{1-r}{r}} (1 - \lambda)\mathcal{M}^{r-1} \\ w_F &= B [(1 - \lambda)\mathcal{M}^r + \lambda\mathcal{F}^r]^{\frac{1-r}{r}} \lambda\mathcal{F}^{r-1} \end{aligned}$$

In counterfactual simulations, wages w_M and w_F change depending on how $\mathcal{M}$ and $\mathcal{F}$ change. This is the key difference from the baseline setup with a linear production function.

The wage ratio is

$$\frac{w_F}{w_M} = \frac{\lambda}{1-\lambda} \left(\frac{\mathcal{M}}{\mathcal{F}} \right)^{1-r} \quad (28)$$

For each gender, normalize, as 1 unit of effective labor, the labor provided by singles with schooling level of 1, market ability of 1, and zero children ($\mathcal{K}_0$), i.e. $\bar{w}_M^{10}(\mathcal{K}_0)$ and $\bar{w}_F^{01}(\mathcal{K}_0)$.

$$\begin{aligned} \mathcal{M} &= \sum_{q=1}^S \sum_{r=0}^S \sum_{\mathcal{K}} n^{qr} d^{qr}(\mathcal{K}) \left(\frac{\bar{w}_M^{qr}(\mathcal{K})}{\bar{w}_M^{10}(\mathcal{K}_0)} \right) \left(P_M^{qr}(\mathcal{K}) \right)^{1-\frac{1}{\theta}} \Gamma \left(1 - \frac{1}{\theta} \right) \\ \mathcal{F} &= \sum_{r=1}^S \sum_{q=0}^S \sum_{\mathcal{K}} n^{qr} d^{qr}(\mathcal{K}) \left(\frac{\bar{w}_F^{qr}(\mathcal{K})}{\bar{w}_F^{01}(\mathcal{K}_0)} \right) \left(P_F^{qr}(\mathcal{K}) \right)^{1-\frac{1}{\theta}} \Gamma \left(1 - \frac{1}{\theta} \right) \end{aligned}$$

To perform counterfactual simulations, I take the following steps. First, the share parameter λ is backed out from equation (28) in the status quo (in status quo, $w_M = \bar{w}_M^{10}(\mathcal{K}_0)$ and $w_F = \bar{w}_F^{01}(\mathcal{K}_0)$):

$$\lambda = \frac{\bar{w}_F^{01}(\mathcal{K}_0) \mathcal{F}^{1-r}}{\bar{w}_F^{01}(\mathcal{K}_0) \mathcal{F}^{1-r} + \bar{w}_M^{10}(\mathcal{K}_0) \mathcal{M}^{1-r}}$$

Next, the exogenous Total Factor Productivity parameter B is backed out from equation (H.2.1):

$$B = \frac{\bar{w}_M^{10}(\mathcal{K}_0)}{[(1-\lambda)\mathcal{M}^r + \lambda\mathcal{F}^r]^{\frac{1-r}{r}} \lambda \mathcal{F}^{r-1}}$$

To solve the model, I iterate until convergence.

1. Start with $w_F^{(0)} = \bar{w}_F^{01}(\mathcal{K}_0)$, $w_M^{(0)} = \bar{w}_M^{10}(\mathcal{K}_0)$.
2. Find $\mathcal{M}^{(1)}$, $\mathcal{F}^{(1)}$ when the wages are $w_F^{(0)}$, $w_M^{(0)}$ in the counterfactual (This coincides with the counterfactual in the linear production function case).
3. Find the new wages $w_F^{(1)}$, $w_M^{(1)}$ that are implied by $\mathcal{M}^{(1)}$, $\mathcal{F}^{(1)}$.
4. Repeat steps 2 and 3 until

$$\left| \left(\frac{w_F}{w_M} \right)^{(n)} - \left(\frac{w_F}{w_M} \right)^{(n-1)} \right| \leq 0.0000001$$

Finally, $Y^{mkt} = w_M^{(n)} \times \mathcal{M}^{(n)} + w_F^{(n)} \times \mathcal{F}^{(n)}$.

H.2.2 Case 3: Model Extension with Income Effects

Case 3 follows the description of the model extension in Appendix D.4, where the final consumption good of a household is produced according to the CES production function with market goods and home production as inputs.

The calculation of aggregate market output remains the same as in the baseline, normalized to the average real hourly wage. Aggregate total output, however, is computed as the average of the final consumption good produced in each household (equation (24)), unlike in the baseline, which simply adds the average real hourly wage and the average real hourly home productivity.

In the counterfactual scenario, married women's optimal labor supply changes with τ , as given by equation (25). The household's economic utility also adjusts accordingly, reflecting the revised budget constraint where the final consumption good now appears on the right-hand-side.

The rest of the counterfactual steps proceed in the usual way, described in Appendix H.1.

H.2.3 Cases 4 and 5: Compression of Wage Differentials

For cases 4 and 5, I compress the observed wage differentials by group in the following way.

- Case 4: The variation in $\bar{w}_g^{qr}(\mathcal{K})$ by marital status and spousal education is suppressed. Denote the new $\bar{w}_g^{qr}(\mathcal{K})$ in case 4 as $\bar{\bar{w}}_g^s(\mathcal{K})$. $\bar{\bar{w}}_g^s(\mathcal{K})$ is set to equal the weighted average of the baseline $\bar{w}_g^{qr}(\mathcal{K})$ for each combination of year, sex, own education level, and family composition, using the empirical share of each group as weights. The same is done for $\bar{h}_M^{qr}(\mathcal{K})$.⁶³
- Case 5: In addition to Case 4, I further scale down the variation in $\bar{w}_g^{qr}(\mathcal{K})$ by own education to 85% of its baseline value. Denote the new $\bar{w}_g^{qr}(\mathcal{K})$ in case 5 as $\tilde{w}_g^s(\mathcal{K})$. I first calculate how much the baseline $\bar{w}_g^{qr}(\mathcal{K})$ increases with own education level on average. To this end, I run a group-level regression of $\bar{w}_g^{qr}(\mathcal{K})$ on a continuous measure of schooling, controlling for year dummies and weighting each group by its empirical share, separately for each gender and family composition:

$$\bar{w}_{jt} = \alpha^{g\mathcal{K}} + \beta^{g\mathcal{K}} \times s_j + \delta_t^{g\mathcal{K}}$$

where $\bar{w}_{jt}$ is the short-hand notation for $\bar{w}_g^{qr}(\mathcal{K})$ of group j with own education level s_j in year t . Then, I start by setting $\tilde{w}_g^1(\mathcal{K}) = \bar{\bar{w}}_g^1(\mathcal{K})$, taken from case 4. Next, I calculate $\tilde{w}_g^s(\mathcal{K}) = \tilde{w}_g^1(\mathcal{K}) + 0.85 \times \hat{\beta}^{g\mathcal{K}} \times (s - 1)$ for $s \geq 2$. The same is done for $\bar{h}_g^{qr}(\mathcal{K})$, the home productivity values of both genders.

⁶³There is no change in women's home productivity $\bar{h}_F^s(\mathcal{K})$, which already does not vary by marital status or spousal education.

Appendix I Wages Gaps by Education: Treatment vs. Selection

The goal of this section is to isolate the role of the treatment effect of acquiring a particular level of education as a driver of the average wage for that level, separate from any selection effects. For this, I use the estimation results in [Chiappori et al. \(2018\)](#).

In [Chiappori et al. \(2018\)](#), $W(\theta_i, s_i, g)$ is the market wage faced by an individual i of gender g , education s_i , and ability θ_i . Individuals of each gender are categorized into 6 types (human capital levels), from the combination of 3 education levels – secondary ($s = 1$), high school ($s = 2$), and university ($s = 3$) – and 2 ability levels – high ($\theta = H$) and low ($\theta = L$). $W(\theta_i, s_i, g)$ is estimated as

$$\ln W(\theta_i, s_i, g) = \alpha^g \cdot \mathbb{1}(\theta_i = L, s_i = 1) + \beta_2^g \cdot \mathbb{1}(s_i = 2) + \beta_3^g \cdot \mathbb{1}(s_i = 3) + \gamma_{s_i}^g \cdot \mathbb{1}(\theta_i = H)$$

where α^g indicates the gender-specific base wage for a low-ability individual with secondary education, i.e. $\ln W(\theta_i = L, s_i = 1, g)$; β_s^g indicates the gender-specific education premium of acquiring education level $s \in \{2, 3\}$; and γ_s^g indicates the gender $\times$ education-specific ability premium for $s \in \{1, 2, 3\}$. The following estimation results are used as ingredients in my calculation.

- Table 7: $\mathbb{P}((\text{husband's type}, \text{wife's type})|\text{married})$
- Table 6: $\mathbb{P}(\text{type}|\text{single}, g)$
- Table 11: $\mathbb{P}(\text{married}|g)$
- Table 9: $\mathbb{P}(s|g)$
- Table 13: Education premiums, ability premiums, base wages, i.e. $\hat{\alpha}^g, \hat{\beta}_2^g, \hat{\beta}_3^g, \hat{\gamma}_1^g, \hat{\gamma}_2^g, \hat{\gamma}_3^g$

I follow the following steps. First, calculate $\mathbb{P}(\text{type}|\text{married}, g)$ from $\mathbb{P}((\text{husband's type}, \text{wife's type})|\text{married})$. Next, the share of types by gender is

$$\mathbb{P}(\text{type}|g) = \mathbb{P}(\text{type}|\text{married}, g) \times \mathbb{P}(\text{married}|g) + \mathbb{P}(\text{type}|\text{single}, g) \times \mathbb{P}(\text{single}|g)$$

I then use $\mathbb{P}(\text{type}|g)$ to calculate, for each gender, the share of high-ability individuals in each education level:

$$\mathbb{P}(H|s, g) = \frac{\mathbb{P}(H, s|g)}{\mathbb{P}(H, s|g) + \mathbb{P}(L, s|g)}$$

The average predicted wage by education for each gender takes into account the education premium, ability premium, and the share of high-ability individuals:

$$\mathbb{E}(\widehat{W}|s, g) = e^{\widehat{\ln W}(H, s, g)} \times \mathbb{P}(H|s, g) + e^{\widehat{\ln W}(L, s, g)} \times \mathbb{P}(L|s, g)$$

Denote as $\tilde{W}$ the predicted wage that includes just the education premium, while excluding the role of ability. Then,

$$\mathbb{E}(\tilde{W}|s, g) = \begin{cases} e^{\hat{\alpha}^g}, & \text{if } s = 1 \\ e^{\hat{\alpha}^g + \hat{\beta}_s^g}, & \text{if } s = 2, 3 \end{cases}$$

Table A9 reports the values of $\mathbb{E}(W|s, g)$ and $\mathbb{E}(\tilde{W}|s, g)$.

Table A9: Wage by Education

Education level	Male		Female	
	Total	Only edu premium	Total	Only edu premium
1	22.68	20.41	15.51	11.60
2	24.17	21.85	15.74	14.84
3	23.89	22.44	20.90	19.11

Note: This table reports two types of wages (earnings, actually, in thousands of 2008 GBP) by gender. “Total” indicates the total wage by education, i.e. $E(W|s, g)$. “Only edu premium” indicates the part of wage that excludes the role of ability, i.e. $E(\tilde{W}|s, g)$.

Finally, Table A10 reports the role of the education premium in accounting for the total wage gap between any two education levels s and s' :

$$\frac{E(\tilde{W}|s', g) - E(\tilde{W}|s, g)}{E(W|s', g) - E(W|s, g)}$$

The education premium accounts for a large share of the total wage gap by education. The smallest (positive) share is 83%. Shares often exceed 100%, which occurs when the ability premium is higher at a lower education level than at a higher one, and/or when the proportion of high-ability individuals is greater among those with lower education than among those with higher education. If the extent of these effects is large, the average wage can actually decline with higher education, which leads to a negative value for the education premium’s share.

Table A10: Education Premium’s Role in Wage Gap by Education

	Compared education levels		
	2 vs. 1	3 vs. 2	3 vs. 1
Male	96%	-209%	168%
Female	1372%	83%	139%

Note: This table reports the share of the wage gap by education that is accounted for by the education premium, for any two education levels.

Comparison with Returns-to-Schooling Literature Interestingly, the results of this exercise mirror empirical estimates of ability bias in the returns to schooling. In the pioneering work of [Griliches \(1977\)](#), adding a control for IQ reduces the OLS estimate of the returns to schooling from 6.8% to 5.9%, suggesting that selection effects account for about 13% of the wage increase associated with an additional year of schooling. This is in a similar range to the compression of educational wage differentials by 15% that I adopt in case 5 of Figure 11 to remove the selection effect of education.

However, [Griliches \(1977\)](#) and many others, as surveyed in [Card \(1995\)](#), find that IV estimates of the returns to schooling typically exceed the OLS estimates. This pattern is consistent with some shares exceeding 100% in Table A10. It also suggests that the education premium might even be larger than the observed wage differentials by education, in which case the fall in output in the main counterfactual would be even larger than in the baseline.